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RESEARCH ARTICLE

SwarmSAR-FW: Drone-Based Swarm Search and Rescue in Fire and Water Disasters

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ABSTRACT Fire and water disasters impose critical constraints on search-and-rescue (SAR) operations through rapidly evolving hazards, degraded visibility, and narrow intervention windows. Conventional SAR operations that rely on human operators and single-UAV approaches have limited scalability under such conditions. To address this, our research introduces SwarmSAR-FW, a coordinated multi-drone framework for integrated fire and flood disaster response that employs cooperative surveying, dynamic task allocation, and real-time communication to enable parallel coverage and adaptive deployment. In aquatic scenarios, the system integrates lightweight wearable devices that provide victim localization, physiological monitoring, drift prediction, and priority-based triage. Autonomous life-jacket deployment achieved visual detection within 2.1 s and a mean delivery accuracy of 1.8 m. The prioritization algorithm correctly ranked critical victims in 98.7% of trials, while the swarm-based task reassignment introduced 4.1 s of coordination latency and 25.4 s of end-to-end assistance time when tested with two physical and one simulated drone. For fire environments, the framework supports multi-altitude detection, panoramic reconstruction, safe access-point identification, and temporally weighted fire localization through aggregated visual observations. Experiments with physical drones using controlled visual injection demonstrate reliable fire-region identification and robust entry/exit estimation in dense smoke. SwarmSAR-FW offers a scalable SAR solution that improves response time, prioritization accuracy, and responder safety relative to single-UAV deployments, thereby addressing fundamental limitations in conventional disaster-response paradigms.

INDEX TERMS Disaster response, search and rescue (SAR), drone swarms, remote sensing, victim detection.

I. INTRODUCTION

The International Association of Fire and Rescue Services (CTIF) estimates that between 100,000 and 120,000 fatalities occur worldwide each year as a direct result of fire-related incidents [1]. Structural and wildland fires remain a major global safety concern, causing substantial loss of life, extensive infrastructure damage, and long-term environmental degradation [2]. Beyond immediate fire-related fatalities, a large number of victims succumb to severe burn injuries and are exposed to smoke, particularly when there are delays in rescue and medical intervention. Burn-related injuries are estimated to account for approximately 180,000 deaths annually worldwide [3], highlighting their substantial global mortality burden. Beyond human casualties, fires also have a significant global economic impact, with annual losses exceeding USD 300 billion from residential fires, industrial incidents, and large-scale wildfires [4]. In recent years, the frequency and severity of extreme wildfire events have increased, affecting millions of acres annually and triggering mass evacuations, particularly in the wildland-urban interface areas.

At the same time, water-related disasters, such as floods and storms, are the deadliest natural hazards worldwide. In 2022, a mega-flood in the Indus River basin of Pakistan submerged nearly 9% of the country, resulting in

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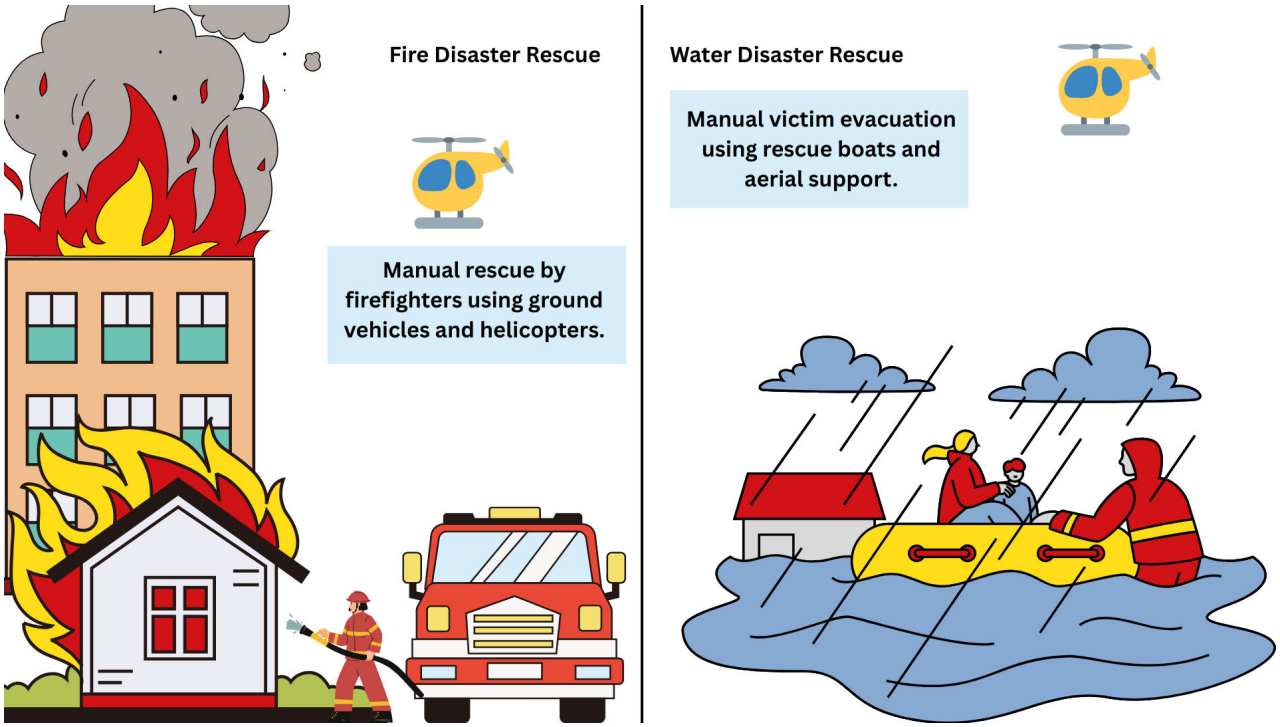


FIGURE 1. Traditional SAR operations highlighting human-led fire and water disaster response using ground teams, boats, and aerial support.

approximately 1,700 deaths, affecting an estimated 33 million people, and displacing nearly 8 million people, with total economic losses exceeding USD 30 billion [5]. During the past five decades, water-related disasters have caused nearly 70% of all fatalities related to natural disasters [6] and more than 90% of major global disaster events since 1970 [7].

Despite the differing physical dynamics of fire and flood scenarios, both disasters share critical operational challenges, including rapidly evolving environments, limited situational awareness, limited evacuation windows, and high risk to victims and first responders alike. The scale, frequency, and severity of these disasters underscore the urgent need for rapid, reliable, and adaptive emergency response mechanisms that can operate effectively across both fire- and water-dominated environments. In such cases, search-and-rescue (SAR) operations play a crucial role in minimizing casualties and delivering immediate medical assistance to affected populations.

During emergency situations, SAR operations involve coordinated efforts to locate, assess, and rescue individuals facing life-threatening conditions [8]. Fire, floods, and drowning events often result in high mortality due to narrow rescue windows, hazardous environments, and limited accessibility. Survival durations vary widely depending on environmental conditions and injury severity, ranging from minutes to several days [9], [10]. Because the probability of successful rescue generally decreases with increasing response time, rapid intervention is critical in SAR operations [11].

Traditional SAR approaches, illustrated in Figure 1, rely primarily on firefighter-led manual operations [12]. In structural fires, responders navigate smoke-filled environments using self-contained breathing apparatus and systematic search patterns. Wildfire SAR emphasizes rapid evacuation and perimeter control, and UAVs are increasingly deployed to improve situational awareness. However, most UAV-based frameworks assume known victim locations and employ simplified models, limiting real-world effectiveness [13]. Victim extraction typically employs stretchers, ladders, or hydraulic platforms, whereas high-rise scenarios require aerial rescue methods such as turntable ladders, rope systems, and helicopters.

In water-related disasters, SAR operations utilize boats, rafts, jet skis, rope systems, and helicopters for rooftop evacuations during severe flooding. Although effective in controlled settings, these approaches face significant limitations during large-scale events. Dense smoke, extreme temperatures, rising water levels, and infrastructure instability severely reduce visibility and accessibility, thereby increasing search duration and operational risk [14]. Secondary hazards including flash floods, structural collapse, and exposure to toxic gases further endanger rescue personnel. As shown in Figure 1, conventional operations remain time-consuming, labor-intensive, and resource-intensive in widespread or rapidly evolving scenarios.

UAV swarms offer a transformative approach to disaster response by enabling coordinated, parallel aerial surveillance in complex environments [11], [15]. Compared to single-drone systems, swarms provide enhanced coverage,

redundancy, and resilience, enabling rapid victim detection in smoke-filled zones and flood-affected areas while reducing responder risk [16]. Through distributed sensing, autonomous task allocation, and inter-drone communication, swarm-based systems reduce detection latency and mitigate single-point failures. However, UAVs deployed in SAR missions must operate under highly dynamic conditions such as strong winds, smoke, and thermal turbulence, which can affect flight stability and mission reliability. Consequently, recent research has explored learning-based wind estimation and stall prediction models to improve UAV robustness and ensure safer autonomous operation in critical SAR deployments [17], [18].

Despite these advantages, existing swarm-based SAR research primarily emphasizes area coverage and visual situational awareness, with limited focus on time-critical rescue support, coordinated victim prioritization, and integrated decision-making in rapidly evolving fire and flood scenarios. Fire dynamics, such as flame spread and smoke propagation, can change within seconds, whereas flood conditions involve rising water levels and drifting victims, necessitating continuous, adaptive monitoring. Current approaches lack unified frameworks that leverage swarm intelligence to support actionable rescue planning, efficient resource allocation, and improved response access in combined fire-water disaster environments.

To address these gaps, this work proposes a coordinated UAV-swarm-based SAR framework, termed SwarmSAR-FW, designed to enhance rapid situational assessment, improve victim-detection efficiency, and support timely rescue operations in both fire and water-disaster scenarios. The framework emphasizes cooperative swarm behavior, efficient coverage strategies, and real-time information sharing to improve operational effectiveness under dynamic and hazardous conditions. Accordingly, the main contributions of this paper are summarized as follows:

- Propose SwarmSAR-FW, a UAV swarm–based search and rescue framework tailored for integrated fire and water disaster response.
- Design a coordinated swarm survey strategy to maximize coverage of disaster-affected areas while minimizing response time.
- Develop a collaborative victim detection and localization framework using onboard sensing and inter-drone communication.
- Demonstrate how swarm-based SAR improves responder safety, operational robustness, and victim detection efficiency compared to single-UAV deployments.

The remainder of this paper is organized as follows. Section II reviews related work on UAV- and swarm-based SAR systems. Section III describes the proposed swarm architecture and operational methodology. Section IV presents experimental evaluation and performance analysis. Section V discusses the key findings, practical implications, and limitations of the proposed framework. Finally, Section VI concludes the paper and outlines directions for future research.

II. RELATED WORKS

Recent advancements in drone-assisted SAR systems have demonstrated the growing potential of UAVs, particularly in fire and water-related emergency situations. Prior research has explored several key directions, including UAV-based sensing and perception, autonomous navigation in hazardous environments, and data-driven victim detection using computer vision and machine learning. Collectively, these efforts show that UAVs can substantially improve situational awareness while reducing risks to human responders. However, most existing frameworks focus on single-UAV deployments or localized monitoring, limiting scalability in large-scale and dynamic disaster scenarios. These limitations underscore the need for coordinated, victim-centric, and swarm-based SAR frameworks.

A. DRONE SWARMS IN SAR OPERATIONS

Recent research has increasingly examined drone swarms as a promising alternative to traditional human-led and single-UAV SAR frameworks [19], [20], [21]. For instance, Meshcheryakov et al. [22] explored swarm systems for skiptracing in SAR missions, showing that coordinated multi-UAV deployments can cover large, complex environments more efficiently than individual drones. Their findings indicate that swarm-based search strategies reduce mission duration and operational costs while maintaining search quality through effective task allocation and cooperative decision-making.

Khan [23] further assesses drone swarm performance using metrics such as coverage efficiency, victim detection accuracy, communication robustness, and adaptability to uncertain environments. Their results show that swarm-based systems survey disaster areas significantly faster than single-UAV approaches and achieve high detection accuracy when integrated with thermal imaging and computer vision techniques.

Afrazi et al. [24] proposed an enhanced density-driven control framework for multi-agent UAV SAR systems that integrates optimal transport theory to improve victim detection efficiency in large-scale disaster environments. Their approach dynamically allocates search effort to high-priority regions while accounting for sensing-range constraints, improving coverage and target-detection performance in simulation studies. However, the framework primarily focuses on search optimization and victim discovery, without addressing victim prioritization, rescue assistance, payload-constrained task reassignment, or disaster-specific operational strategies. In contrast, SwarmSAR-FW extends beyond coverage optimization by integrating cooperative victim servicing, wearable-assisted prioritization, dynamic task redistribution, and dedicated workflows for both fire- and water-disaster scenarios.

Al Amin et al. [25] proposed a swarm-based UAV reconnaissance framework using a Gravitational Search Algorithm (GSA) for cooperative target detection in unknown environments. While their approach improves search efficiency and coverage, it focuses primarily on target discovery and reconnaissance. In contrast, SwarmSAR-FW extends swarm coordination to victim prioritization, drift-aware localization, and coordinated rescue assistance for fire and flood disasters.

Beyond technical performance, Hoang et al. [26] investigate real-world deployment challenges through stakeholder interviews and user studies. Their work underscores the operational advantages of swarm coordination while identifying key practical concerns, including human–swarm interaction, interface design, and trust in autonomous systems. Collectively, these studies demonstrate that drone swarms can substantially enhance SAR efficiency, responder safety, and situational awareness, while also highlighting critical challenges that must be addressed to enable widespread adoption.

Complementing swarm-focused research, recent surveys on UAV deployment in emergency settings further validate the expanding role of autonomous aerial systems in disaster response. Herdel et al. [27] identified emergency response as the most prominent domain within human–drone interaction research, emphasizing rapid deployment and reduced human exposure to hazardous environments. Prior studies have demonstrated the effectiveness of UAVs in SAR missions across diverse terrains and conditions [28], [29], [30]. A comprehensive survey by Lyu et al. [31] synthesizes recent advances in UAV-enabled SAR operations, highlighting how advances in sensor integration, autonomous navigation, task assignment, and path planning have collectively enhanced real-time monitoring, victim localization, and operational effectiveness across diverse environments, from dense forests to maritime settings. This review underscores drones' ability to improve mission success rates, situational awareness, and decision support, while also outlining key challenges in automation, data processing, and cooperative control that continue to drive research frontiers.

B. UAV-BASED SAR FOR FIRE DISASTERS

UAVs have become increasingly important in fire disaster monitoring and response due to their rapid deployment, high maneuverability, and ability to operate in hazardous environments. For example, Partheepan et al. [32] presented a comprehensive review of UAV-based bushfire management systems, highlighting advances in AI-driven perception, autonomous navigation, and multi-sensor platforms integrating visual, infrared, and LiDAR data. Their study demonstrates the effectiveness of UAVs for fire detection, prediction, and SAR tasks, while noting that rapidly evolving fire fronts require faster and more scalable response mechanisms.

Building on fine-grained fire detection, Hu et al. [33] proposed a UAV-based monitoring framework supported by a multimodal dataset that combines RGB and thermal imagery. Their results show that lightweight deep learning models can achieve accurate real-time flame detection, segmentation, and tracking on edge-enabled UAV platforms. However, the framework is primarily designed for single-UAV operation. In large-scale or fast-spreading fires where smoke density and fire intensity can change within seconds, reliance on a single drone significantly constrains coverage, response speed, and victim reach [34].

Differently, Rajagopal et al. [35] introduced a hybrid wildfire monitoring framework that integrates wide-area synthetic aperture radar sensing with UAV-based optical and thermal imagery. By leveraging transformer-based architectures and vision–language models, their approach improves fire segmentation under low-visibility conditions and produces human-interpretable insights. While this system enhances situational awareness and early detection, it primarily targets large-scale monitoring rather than time-critical rescue operations or coordinated multi-UAV intervention.

Recent comprehensive surveys provide a broader perspective on UAV applications in wildfire management and SAR. A recent review synthesizes UAS-based remote sensing applications across pre-fire assessment, active-fire surveillance, and post-fire impact analysis, examining studies from major databases such as ScienceDirect, IEEE Xplore, and Scopus [36]. This survey highlights current capabilities and identifies key gaps in onboard AI processing, vegetation-specific fuel estimation, and field validation of navigation methods.

Another recent survey focuses on fire-oriented UAV systems, classifying advances in sensor fusion, autonomous control, and learning-based fire detection, and evaluating their performance under challenging conditions such as dense smoke and rugged terrain [37]. Beyond benchmarking existing algorithms, the survey discusses emerging trends in real-time edge intelligence, cooperative aerial sensing, and adaptive flight planning for dynamic fire environments, emphasizing the need for scalable, resilient UAV architectures for emergency response.

Overall, existing UAV-based research in fire disaster response confirms the effectiveness of drones for situational awareness, fire detection, and risk assessment. Nevertheless, the rapid propagation and dynamic nature of fires render single-drone deployments insufficient in widespread or time-critical scenarios. Limited attention has been paid to swarm-level coordination for real-time response, motivating the need for multi-UAV frameworks capable of parallel coverage, rapid decision-making, and actionable rescue planning.

C. UAV-BASED SAR FOR WATER DISASTERS

Flood disasters create highly time-critical SAR scenarios in which fast-moving water, floating debris, and limited accessibility severely constrain human-led rescue operations [38], [39], [40]. As a result, UAVs have emerged as practical tools for rapid situational awareness and victim detection in

flooded environments. In this line, Gaur et al. [41] proposed a UAV-assisted SAR system employing YOLOv4 for real-time human detection in flood scenarios, achieving improved detection speed and an average accuracy of 79.46%. While effective, this framework relies solely on a single UAV and visual sensing, which limits coverage and responsiveness in large-scale floods where victims may be widely dispersed or drifting with water currents.

Differently, Thuan et al. [42] introduced an AI-enabled UAV platform that integrates deep learning–based victim detection with autonomous payload delivery for flood rescue. Their system demonstrates the feasibility of rapid victim localization and life-saving supply delivery in hazardous conditions. However, reliance on single-drone operation limits scalability and parallel victim assistance, particularly in dynamic flood environments where water currents can quickly shift victims' locations.

A comprehensive review by Minh et al. [43], covering 69 studies across pre-flood, during-flood, and post-flood phases, highlights the extensive use of UAVs for flood mapping, damage assessment, and SAR support. Despite their rapid deployment and operational flexibility, the review identifies persistent limitations, including restricted flight endurance, communication constraints, and the lack of coordinated multi-UAV strategies during active rescue operations.

UAVs have also been integrated with multi-sensor data for high-resolution flood assessment. A recent study by [44] reports on a UAV-mounted LiDAR and RGB multispectral sensor suite deployed during a flash flood event in Spain, enabling rapid topographic mapping, damage assessment, and in-depth environmental analysis that are crucial for SAR planning. Their approach also enables near-real-time detection of terrain changes, supporting adaptive SAR route planning and the safer deployment of rescue personnel. Fawakherji and Hashemi-Beni [45] further enhance flood mapping by integrating UAV optical imagery with satellite SAR data, achieving high-precision inundation maps that can better guide emergency response and resource deployment. This multi-platform fusion also improves the detection of hidden flood zones beneath vegetation or debris, reducing the risk of overlooking hazard areas during rescue operations. In addition, Zhao et al. [46] propose a real-time UAV-based flood-rescue target-detection method that efficiently identifies people and vehicles trapped by floodwaters, thereby improving situational awareness and facilitating rapid SAR decision-making. Moreover, their system incorporates dynamic water-flow modeling to predict victim drift, thereby enabling preemptive UAV positioning for faster victim localization.

In summary, existing UAV-based approaches for flood disasters confirm the effectiveness of drones in enabling rapid victim detection, real-time monitoring, and safe supply delivery. Nevertheless, the highly dynamic nature of floods and the mobility of victims render single-UAV systems insufficient for timely and comprehensive rescue. Critical challenges including cooperative area partitioning, victim drift tracking, and prioritized assistance under time and payload constraints remain largely unexplored, underscoring the need for coordinated multi-UAV SAR frameworks.

D. SUMMARY

Table 1 summarizes the evolution of UAV-based search-and-rescue methods, from classical single-UAV systems to coordinated drone swarms, and highlights the principal contribution and remaining limitation of each methodological category. Drone-based SAR systems face a fundamental trade-off between sensing capability, payload weight, and operational endurance. Advanced sensors such as LiDAR and thermal cameras enhance perception in low-visibility environments but increase power consumption and reduce flight time. In contrast, lightweight RGB cameras maintain endurance but lack reliability for victim detection and prioritization in complex fire and flood scenarios. This necessitates multimodal sensing strategies that balance perception quality with flight performance. Most existing UAV-based SAR frameworks rely on single-drone, perception-centric approaches suited for localized monitoring. However, they scale poorly in large, dynamic disasters where victims may be dispersed, drifting, or obscured. Additionally, limited integration of real-time victim assessment and coordinated task allocation restricts prioritization and operational efficiency, while continued reliance on manual intervention exposes responders to unnecessary risk.

To address these challenges, we propose SwarmSAR-FW, a drone-based swarm SAR framework tailored to fire and water-disaster scenarios. The proposed system integrates cooperative multi-UAV navigation, lightweight wearable victim bands for real-time localization and health monitoring, and a centralized coordination dashboard to enable parallel search, dynamic victim prioritization, and rapid response. By combining swarm intelligence with victim-centric sensing, SwarmSAR-FW enhances area coverage, reduces response time, and improves rescue planning in highly dynamic and hazardous disaster environments.

III. METHODOLOGY

This section describes the proposed drone-assisted rescue framework (SwarmSAR-FW), detailing the hardware architecture, cooperative survey strategy, victim detection and prioritization mechanisms, and the integrated drone–cloud communication system. Table 2 summarizes the key notation used in survey partitioning, fire likelihood estimation, region scoring, and drone coordination.

A. SYSTEM HARDWARE ARCHITECTURE

This subsection describes the hardware design of the proposed system, including the aerial drone platform and the wearable victim detection device.

TABLE 1. Representative UAV-based and drone-swarm approaches for fire and water disaster search and rescue.

Approach	Representative Work	Core Idea / Contribution	Relevance and Remaining Gap
UAV-Based Fire Monitoring	Hu et al. [33], Partheepan et al. [32]	UAVs are investigated for fire and bushfire monitoring using aerial remote sensing, including visible/infrared sensing, AI-based detection, and autonomous aerial observation.	These studies primarily address fire detection, monitoring, and management rather than direct victim search, rescue assistance, or coordinated victim response.
Single-UAV Flood SAR	Gaur et al. [41], Thuan et al. [42]	Gaur et al. use UAV imagery and YOLOv4 for human detection in flood SAR, while Thuan et al. develop a drone-based rescue platform that combines intelligent human detection with a simulated multi-payload delivery mechanism.	These studies demonstrate the use of individual UAVs for flood victim detection and rescue support, but do not address coordinated multi-UAV search or simultaneous assistance to multiple victims.
Real-Time Flood Target Detection	Zhao et al. [46]	Develops a UAV-based method for real-time flood extent extraction and detection of rescue targets, including pedestrians and vehicles, from flood imagery.	Addresses flood perception and rescue-target detection rather than coordinated multi-UAV search, victim prioritization, dynamic victim tracking, or physical rescue-payload deployment.
UAV-Based Flood Assessment	Roman et al. [44]	Uses UAV-mounted sensing during a real flash-flood event for three-dimensional reconstruction, multispectral assessment, vehicle and debris detection, gas sensing, and flood-forecasting support.	Provides detailed environmental and damage assessment for emergency response, but does not constitute an integrated victim-search and rescue framework.
UAV–Satellite Flood Sensor Fusion	Fawakherji and Hashemi-Beni [45]	Combines high-resolution UAV optical imagery with Sentinel-1 satellite SAR data to improve flood inundation mapping, including assessment of inundated and dry vegetation in complex environments.	Addresses flood mapping through multi-platform sensor fusion rather than direct victim localization, rescue assistance, or multi-UAV coordination.
Drone-Swarm SAR: Opportunities and Challenges	Hoang et al. [26]	Reviews the potential application of drone swarms to SAR and discusses challenges related to situational awareness, technical requirements, human–drone interaction, and operational adoption.	Provides a conceptual perspective on swarm-based SAR rather than an integrated experimental system combining autonomous search, victim prioritization, and physical rescue assistance.
Swarm-Based Missing-Person Search	Meshcheryakov et al. [22]	Investigates the application of a swarm of quadcopters for searching for missing persons, demonstrating the potential of distributed UAV deployment for search operations.	Addresses distributed aerial search but does not integrate flood-specific victim dynamics, rescue-payload delivery, or priority-based victim assistance.
Drone-Swarm SAR	Khan [23]	Investigates drone-swarm deployment for SAR, considering search efficiency, victim detection, communication resilience, energy utilization, and adaptability.	Demonstrates the potential of swarm-based SAR but does not address the integrated combination of fire and flood response, victim prioritization, dynamic task reassignment, and physical rescue-payload deployment considered in this work.

1) AERIAL PLATFORM

A quadcopter drone powered by lithium-polymer (LiPo) batteries serves as the aerial platform. The drone is equipped with a Pixhawk flight controller, a Global Positioning System (GPS) module, and a telemetry module, as illustrated in Figure 2. The Pixhawk flight controller is the central control unit that handles low-level flight stabilization, waypoint navigation, and sensor integration. The GPS module provides satellite-based positioning for coordinate-driven navigation. The telemetry module enables bidirectional communication between the drone and the ground control station for mission monitoring and control.

In addition to the flight controller, the drone integrates a Raspberry Pi single-board computer and an onboard camera. The Raspberry Pi performs high-level computation, mission logic, and communication with the flight controller, issuing control commands when required. The onboard camera is used for detecting humans, fire, smoke, and open spaces suitable for rescue entry and exit. An external Wi-Fi module provides wireless connectivity, enabling internet access and communication with ground infrastructure and wearable victim devices during missions.

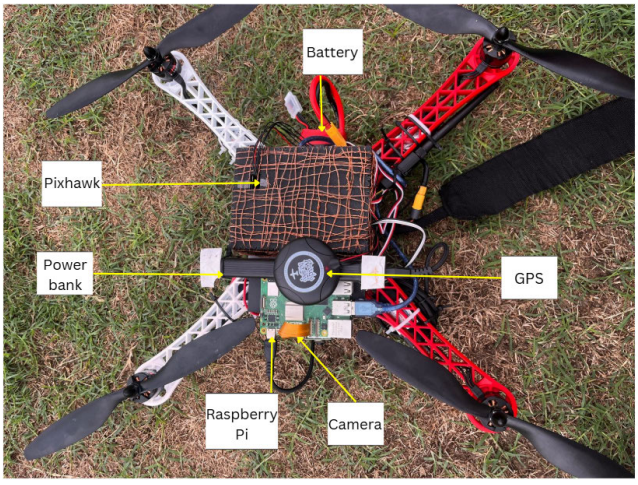


FIGURE 2. Hardware configuration of the quadcopter.

TABLE 2. Key notation used in the proposed algorithms.

Symbol	Description
TL, TR, BL, BR	Corner coordinates of the surveyed disaster region
$\mathcal{A}$	Convex aerial survey area
c_i	i-th partitioned grid cell within $\mathcal{A}$
$D_{\max}$	Maximum diagonal length of each cell
R	Drone communication/sensing range
$d_{\max}$	Maximum drone–target separation within a cell
$\mathbf{p}_0$	Initial drone position
$\mathbf{p}_{\text{entry}}$	Nearest entry point into a grid cell
r	Circular inspection radius inside a cell
$I_{\text{fire}}(x, y, t)$	Binary indicator of fire or smoke detection at pixel (x, y) and time t
$\text{FireDensity}(x, y)$	Spatial accumulation of fire observations
$t_{\text{first}}(x, y)$	First detection time of fire at location (x, y)
$\text{FireScore}(x, y)$	Temporally weighted fire likelihood score
$\mathcal{R}_k$	Connected fire candidate region k
S_k	Cumulative FireScore of region $\mathcal{R}_k$
d_k	Euclidean distance between drone and region $\mathcal{R}_k$
Priority_k	Priority value of region $\mathcal{R}_k$
α, β	Fire intensity and distance weighting factors
$\vec{v} = (v_x, v_y)$	Water flow or drift velocity vector
Δt	Time interval between localization and response
(x_p, y_p)	Predicted displaced target position
$\mathcal{D}$	Set of drones in the swarm
P_i	Remaining payload/energy of drone i
S_i	Operational state of drone i

2) WEARABLE VICTIM DEVICE

The wearable victim device is built around an ESP32 microcontroller, as shown in Figure 3, which acts as the primary processing and communication unit. The device continuously collects sensory data and transmits it to a cloud-based Firebase database for real-time access by the rescue system.

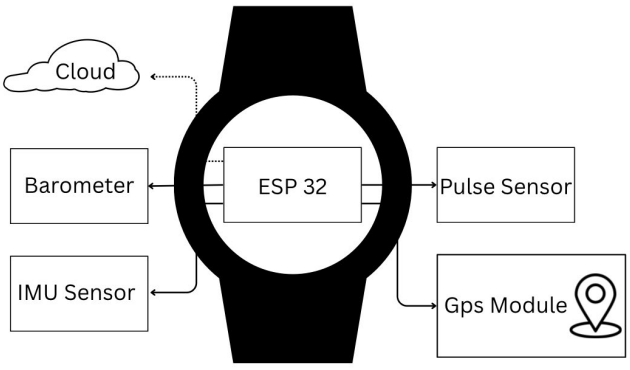


FIGURE 3. Wearable device configuration based on the ESP32 microcontroller, integrating the IMU, GPS, and vital sign monitoring.

To ensure robustness in adverse conditions, the wearable employs sensory fusion using a GPS module, an inertial measurement unit (IMU), a waterproof barometric sensor, and a pulse sensor. The GPS module provides geographic location, while the barometric sensor estimates altitude and supports operation in flood environments. The IMU enables inertial dead reckoning and supports motion and drift estimation when GPS signals are degraded. In GPS-denied environments, the wearable device retains the last valid GPS position as a dead-reckoning anchor, while the onboard IMU continuously estimates the victim's relative movement. The estimated position is subsequently used to guide rescue operations. In the current implementation, autonomous UAV navigation still relies on GPS-assisted waypoint navigation; therefore, complete GPS-denied UAV operation is beyond the scope of this work. Nevertheless, the proposed swarm coordination and task allocation framework is independent of the underlying localization technology and can be integrated with alternative localization methods, such as vision-based SLAM, LiDAR-SLAM, or ultra-wideband (UWB) positioning, enabling future deployment in GPS-denied environments without modifying the coordination framework. The pulse sensor measures heart rate, serving as a physiological indicator for assessing the victim's condition and prioritizing care. The wearable device incorporates a compact GPS module that provides meter-level positioning under typical operating conditions. In the current experimental setup, wearable coordinates were initialized using the drone's onboard GPS prior to mission execution. Therefore, the localization error of the wearable GPS was not independently characterized. Consequently, the reported 1.8 m payload deployment accuracy reflects the cumulative error of the entire rescue pipeline, including positioning, drift prediction, navigation, and payload deployment, rather than the wearable GPS localization error alone. Future work will independently evaluate wearable localization accuracy.

Rather than transmitting raw sensor streams, the wearable transmits compact, 90 s summary data blocks. These summaries capture medical trends (such as heart rate statistics, variability, deterioration patterns, and flatline detection), drowning severity (including mean altitude and altitude trends), movement state (IMU-based activity levels), and location stability (GPS position and drift). A triage algorithm processes these temporal blocks to assign weighted priority scores based on medical urgency, drowning risk, mobility, and environmental factors. The system ranks victims accordingly, enabling noise-resistant and resource-efficient rescue decision-making.

Table 3 summarizes the hardware components along with their key configurations and intended purposes. The first section of the table describes the drone setup in detail, while the second section outlines the wearable device.

B. DRONE SEARCH AREA PARTITIONING AND COOPERATIVE SURVEY STRATEGY

The disaster survey region is defined by the user using four geographic corner coordinates, top-left (TL), top-right (TR), bottom-left (BL), and bottom-right (BR), which form

TABLE 3. Hardware components used in the proposed SAR system.

Category	Device/Module	Key Specifications	Purpose
Drone Components			
Drone Platform	Custom Quadcopter	Thrust-to-weight ratio ≈ 2:1; Li-Po battery powered	Aerial survey and victim detection
Flight Controller	Pixhawk 2.8	Integrated Inertial Measurement Unit (IMU), barometer; supports autonomous navigation	Flight stabilization and autonomous control
Onboard Computer	Raspberry Pi 5	Quad-core 2.4 GHz, 8 GB RAM; serial communication with Pixhawk	Onboard computation, data processing, and communication hub
Camera	Camera Module with Servo Gimbal	Pan-tilt capability for dynamic FOV	Visual sensing for victim detection and environment mapping
Communication	Telemetry with Wi-Fi Module	Two-way link with ground station; Wi-Fi powered independently	Data transmission and network connectivity
Drone GPS	External GPS Module	Provides latitude/longitude geolocation	Accurate drone positioning and victim localization reference
Wearable Device Components			
Wearable MCU	ESP32	Dual-core 240 MHz, Wi-Fi + BLE, low power	Controller for all wearable sensors and communication with drone
Vital Sign Sensor	Pulse Sensor	Analog BPM output	Victim triage based on heart rate
Motion/Orientation Sensor	IMU	3-axis accelerometer + gyroscope	Victim motion monitoring and dead-reckoning localization
Altitude Sensor	Barometer (water resistant)	Atmospheric pressure-based altitude estimation	Victim vertical positioning
Wearable GPS	Compact GPS Module	UART interface, meter-level accuracy	Victim geolocation

a convex quadrilateral operational area. Upon receiving these boundaries, the system partitions the region into smaller, non-overlapping square cells, as presented in Algorithm 1.

Here, cell dimensions are determined by the number of available drones, N, and the effective wireless communication range, $R = 50m$. Each cell is constrained such that its diagonal length must be $D_{max} = 150m$, as illustrated in Figures 4 and 5. This ensures that the maximum drone–victim separation during circular traversal remains bounded. Specifically, $d_{\max}$ represents the maximum radial distance from the circular survey trajectory to the outermost point of the cell, estimated using Eq. 1. This value ensures reliable wireless connectivity and timely data exchange.

$$d_{\max} = \frac{D_{\max}}{2} - R = \frac{150}{2} - 50 = 25 \text{ m} \tag{1}$$

Each cell is assigned to a single drone, enabling parallel coverage and significantly reducing mission completion time, which is essential for time-critical rescue operations. Once assigned, each drone travels from its initial position $\mathbf{p}_0$ to the nearest boundary point, as defined in Eq. 2,

$$\mathbf{p}_{\text{entry}} = \arg\min_{\mathbf{p}\in\partial c_i} \|\mathbf{p} - \mathbf{p}_0\| \tag{2}$$

where $\mathbf{p}$ denotes an arbitrary point on the boundary ∂c_i of the grid cell, and p_{entry} represents the closest such point serving as the drone's entry location. From this point, the drone initiates a circular survey pattern. The radius of the circular trajectory is maintained using Eq. 3.

$$r = R = 50 \text{ m}, \tag{3}$$

corresponding to the maximum reliable Wi-Fi communication range. This ensures that at any point during the survey,

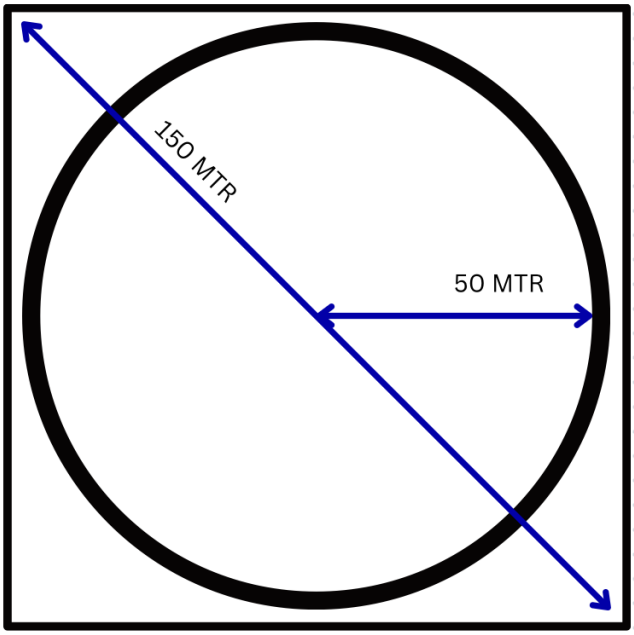


FIGURE 4. Geometric constraints used for search area partitioning, illustrating the maximum allowable cell dimensions and diagonal length to ensure bounded drone–victim distance during circular survey.

victims within the cell satisfy the constraint presented in Eq. 4.

$$\|\mathbf{p}_{\text{victim}} - \mathbf{p}_{\text{drone}}\| \leq R \tag{4}$$

and remain connected to the drone for continuous data exchange. The circular motion provides uniform coverage of the entire cell while minimizing sharp maneuvers, thereby improving flight stability and reducing energy consumption.

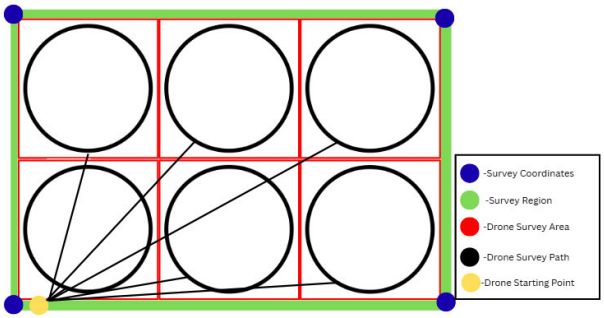


FIGURE 5. Partitioned survey layout showing assignment of square cells to individual drones and the corresponding circular survey trajectories used to maintain reliable communication coverage.

Algorithm 1 Search Area Partitioning and Cooperative Circular Survey

Require: Boundary coordinates TL, TR, BL, BR, number of drones N, communication range $R = 50$ m
Ensure: Complete coverage of the disaster search area

```
Construct convex search region A using {TL, TR, BL, BR}
Compute maximum cell side length s ← 150/√2
Partition region A into non-overlapping square cells C = {c_1, c_2, ..., c_M} with side length ≤ s
Assign each cell c_i ∈ C to a unique drone d_i, i ≤ min(M, N)
for each assigned drone d_i do
    Compute closest entry point p_i on the boundary of cell c_i from the starting location of d_i
    Navigate drone d_i to entry point p_i
    Compute cell center o_i of c_i
    Initiate circular survey trajectory centered at o_i with radius R
    while survey of c_i not completed do
        Capture sensor data and scan for victims
        Maintain wireless communication with detected victims
    end while
    Mark cell c_i as surveyed
end for
return Survey completion status = 0
```

C. VICTIM DETECTION AND ASSISTANCE IN DYNAMIC WATER-BASED DISASTERS

In this context, water-based disasters refer to emergencies such as floods and storm-induced inundations, in which victims may be partially or fully submerged in moving water and are continuously displaced by water currents. The proposed framework is designed to operate in dynamic aquatic environments, where victim locations change over time as a function of flow velocity and surrounding environmental conditions.

Each victim is equipped with a lightweight wearable device that transmits essential physiological and positional data, including geographic coordinates (latitude and longitude), altitude, and pulse rate. These data are periodically uploaded to a centralized database and made accessible to the rescue drone both prior to and during the mission. The drone is equipped with a GPS module, an onboard camera, and a payload deployment mechanism for life jackets. Upon deployment, the drone conducts a predefined aerial survey of the assigned flood-affected area. During this phase, the system identifies all victims within the survey region using reported wearable device data and assigns them to the drone for individual inspection.

1) VICTIM DRIFT PREDICTION IN FLOOD CONDITIONS

In flood scenarios, victims are subject to continuous displacement due to water flow. Since there exists a time delay between the last reported band position and the drone's arrival, the victim's current location must be predicted to ensure accurate interception.

Let the last known victim position be x_0, y_0 recorded at time t_0. Assume that the drone arrives in the victim's vicinity at time t_1. The elapsed time is computed as shown in Eq. 5:

$$\Delta t = t_1 - t_0. \tag{5}$$

Assuming the victim drifts with the local water current velocity as shown in Eq. 6:

$$\vec{v} = (v_x, v_y), \tag{6}$$

the predicted victim position is computed as shown in Eqs. 7–8:

$$x_p = x_0 + v_x \cdot \Delta t \tag{7}$$

$$y_p = y_0 + v_y \cdot \Delta t \tag{8}$$

Here, v_x and v_y represent the horizontal components of the estimated water flow velocity, obtained either from prior environmental data or inferred from consecutive victim position updates.

2) SEARCH OFFSET AND VISUAL DETECTION STRATEGY

To compensate for uncertainties in drift estimation and sensor noise, the drone does not directly navigate to the predicted point (x_p, y_p). Instead, it travels an additional offset distance of approximately 5 m downstream of the predicted location. From this offset point, the drone initiates a systematic visual scan upstream. This upstream scanning strategy increases the probability of detecting the victim within the camera's field of view, even when the drift velocity has been underestimated or the victim encounters obstacles.

The accuracy of drift prediction depends on the quality of the estimated drift velocity. To improve robustness against estimation errors, the drone applies a 5 m downstream offset before initiating visual search, followed by an upstream scan around the predicted location. This search strategy enlarges the effective camera search region, enabling recovery from moderate velocity estimation errors. Future work will include a comprehensive quantitative sensitivity analysis of drift estimation errors.

3) SWARM-BASED COOPERATIVE RESCUE AND DYNAMIC TASK REASSIGNMENT

Once a victim is visually detected through the onboard camera feed, the drone stabilizes its position to verify the victim's presence. Human detection is performed using the YOLOv11n model. We selected YOLOv11n because the system's primary objective is real-time victim localization rather than maximizing benchmark detection accuracy. Larger detectors may provide marginal accuracy improvements but incur substantially higher inference latency and computational requirements. The lightweight YOLOv11n architecture enables near-real-time processing on the deployed Raspberry Pi edge platform while maintaining adequate detection performance for rescue operations. On the deployed Raspberry Pi 5, YOLOv11n achieves an average inference time of 78 ms per frame (approximately 12.8 FPS), enabling near-real-time processing. This enables real-time identification of people in aerial imagery. Upon confirmation, the drone autonomously deploys a life jacket in close proximity to the detected victim. Simultaneously, the victim's status in the centralized database is updated to indicate successful assistance. To reduce duplicate victim assignments, the centralized database maintains a temporary reservation state for each victim. When a drone is assigned to assist a victim, the victim record is marked as reserved and associated with the assigned drone. Prior to initiating the rescue action, the drone verifies the reservation status in the database. If the assignment remains valid, the rescue proceeds; otherwise, the task is released and reassigned accordingly.

The water rescue mission is designed under strict payload and time constraints. Each drone can carry a maximum of three life jackets, a restriction imposed by the drone's thrust-to-weight (T/W) ratio, which directly affects flight stability, maneuverability, and endurance. In addition, the time-critical nature of flood rescue operations necessitates rapid response and efficient resource utilization.

When a drone exhausts its available life jackets or when the number of detected victims within its assigned survey region exceeds its carrying capacity, it generates a help request to the centralized coordination system, as illustrated in Figure 6. This request includes the victim's predicted location, urgency level, and the requesting drone's current payload status. The coordination system subsequently activates a swarm-based cooperative rescue mechanism that dynamically reallocates nearby drones to ensure continuous and timely assistance for victims.

Task reassignment within the swarm follows a priority-driven selection strategy as presented in Algorithm 2. The task assignment algorithm has a computational complexity of O(N), where N denotes the number of candidate drones, indicating that computational overhead increases linearly with swarm size. The measured average task reassignment latency of 4.1 s was obtained using two physical drones and one simulated drone, and includes communication, database, and state-update delays in addition to the task-assignment computation. Mission Planner simulations further demonstrate cooperative area partitioning for up to ten drones. In practice, the maximum swarm size is expected to be limited primarily by communication infrastructure, including Firebase concurrent connection limits and wireless network capacity, rather than by the computational complexity of the proposed algorithm. Candidate drones are first filtered by availability, ensuring that only drones not currently engaged in payload deployment or critical victim inspection are considered. Among the available drones, priority is given to those that still carry at least one life jacket, thereby avoiding unnecessary return-to-base operations. If multiple eligible drones are available, the drone with the minimum Euclidean distance to the victim's predicted location is selected to minimize response time. If no drone with an available payload is in the vicinity, the closest drone is instructed to return to the base station for rapid payload replenishment, while the other drones continue surveillance and victim monitoring.

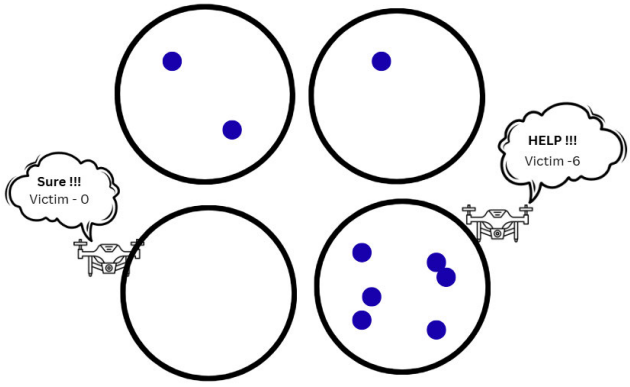


FIGURE 6. Illustration of swarm-based help request generation when a drone exceeds its payload capacity, triggering dynamic reassignment of nearby drones for cooperative victim rescue.

This cooperative decision-making framework allows the swarm to adapt dynamically to evolving rescue demands, balance workload across drones, and minimize idle time. When multiple drones report conflicting observations, the centralized database maintains the most recently validated mission state. All subsequent task-allocation decisions are therefore performed using a common synchronized mission view, reducing inconsistencies among swarm members. The drone then proceeds to the next assigned victim within the survey region until it assists all detectable victims, exhausts its payload capacity, or reaches mission constraints such as battery level and time thresholds.

4) SWARM COORDINATION EVALUATION PROTOCOL

During each rescue mission, the mission controller continuously logged UAV position, payload availability, assigned victims, task completion status, and communication timestamps. Whenever a UAV could no longer service an assigned victim because of payload exhaustion or workload

Algorithm 2 Swarm-Based Dynamic Victim Assignment

Require: Set of drones $\mathcal{D}$, detected victim v, payload status P_i, drone states S_i, drone positions $\mathbf{p}_i$

Ensure: Assignment of an assisting drone

Generate help request with victim location $\mathbf{p}_v$ and urgency level

Identify candidate drones:

$$\mathcal{D}_c = \{d_i \in \mathcal{D} \mid S_i = \text{free or non-critical}\}$$

Filter drones with available payload:

$$\mathcal{D}_p = \{d_i \in \mathcal{D}_c \mid P_i > 0\}$$

if $\mathcal{D}_p \neq \emptyset$ **then**

  Select drone:

$$d^* = \arg\min_{d_i \in \mathcal{D}_p} \|\mathbf{p}_i - \mathbf{p}_v\|$$

else

  Select closest drone for payload replenishment:

$$d^* = \arg\min_{d_i \in \mathcal{D}_c} \|\mathbf{p}_i - \mathbf{p}_v\|$$

  Command d^* to return to base for payload refill

end if

Assign victim v to drone d^*

Update swarm task allocation and drone states

return Assignment status = 0

redistribution, the system automatically generated a help request and forwarded it to the centralized task-allocation module. The coordinator then selected the assisting UAV using the same multi-criteria allocation strategy described in Algorithm 2, considering remaining payload capacity, Euclidean distance to the pending victim, and current mission workload.

For analysis purposes, each reassignment event was categorized according to the requesting and responding platforms (Physical→Physical or Physical→Simulated). Decision latency was computed as the interval between help-request generation and acceptance by the assisting UAV, while assisting UAV arrival time was measured from task acceptance until arrival at the reassigned victim location. These metrics were recorded for every reassignment event during the experimental trials.

5) VICTIM PRIORITIZATION STRATEGY

In water-rescue missions, not all victims require immediate intervention. Some individuals may be located on the water surface in relatively safe regions, such as elevated terrain, rooftops, or floating structures, whereas others may be partially submerged or actively drowning. Given the limited availability of rescue resources and the need to minimize response time in life-threatening situations, an effective victim-prioritization strategy is essential. Therefore, each victim is equipped with a wearable band (discussed in Section III-A2) that integrates a waterproof barometric sensor and a pulse sensor. The barometric sensor provides altitude relative to the surrounding environment, enabling estimation of the victim's position relative to the local water level. This information is used to distinguish victims above the water surface from those at or below it, with the latter group considered at substantially higher risk.

The prioritization procedure is conducted in two sequential stages. In the first stage, victims are classified based on their relative vertical position. Individuals detected at or below the estimated water level are assigned the highest priority, as they are more likely to be drowning or experiencing restricted mobility due to water currents. Conversely, victims detected above the water level are assigned a lower priority, as they may be located on elevated ground, rooftops, or other structures and are therefore less likely to require immediate assistance.

In the second stage, the physiological condition is assessed using pulse-rate measurements. Among victims identified at the surface or below the water level, those exhibiting abnormal or critical pulse values are prioritized for further evaluation, as such readings may indicate severe physical distress or imminent health risks. This ordering assumes that individuals on the water surface face a lower immediate risk of drowning and may remain temporarily safe, for example, on elevated surfaces or while passively drifting with the water flow.

Overall, this two-stage prioritization strategy enables the drone swarm to allocate rescue effort to the most critical victims first, thereby improving response efficiency and mission effectiveness under strict time and payload constraints.

D. VICTIM DETECTION AND ASSISTANCE IN DYNAMIC FIRE-BASED DISASTERS

In rescue operations during fire events, the primary objective of the drone system is to detect human presence and identify the safest possible entry and exit points for rescue personnel. In contrast to flood scenarios, fire environments are characterized by highly dynamic hazards, including variable smoke density, flame propagation, and thermal constraints. These conditions require adaptive surveying strategies and cooperative behavior among multiple drones to ensure rapid and reliable situational awareness. Therefore, a drone dynamically assists neighboring drones after completing its survey of the assigned region when their survey or analysis tasks are still ongoing. This cooperative behavior was evaluated experimentally using two physical drones and one simulated drone instance operating concurrently across separate survey regions. This reduces overall mission completion time and improves coverage reliability, both of which are essential in time-critical fire-rescue operations.

Human detection surveys are conducted at three discrete altitude levels relative to the detected human position: 10 m, 5 m, and 0 m above the human altitude. This multi-altitude

scanning strategy improves robustness to occlusions from smoke, flames, and structural elements while providing multiple viewing perspectives for reliable human detection. At each altitude, onboard visual sensors detect human presence, identify active fire regions, and estimate local smoke concentration.

In parallel with human detection, the drone aims to determine a feasible and safe entry/exit point that enables rescue personnel to access the identified individual. During localized surveying, the drone captures images at four uniformly spaced angular positions along a circular flight path around the target area. This process is repeated across the three altitude levels, yielding a total of 12 images per detected victim (four viewpoints per altitude). The system then processes and stitches these images to generate a panoramic representation of the surrounding environment. Using the resulting panoramic view, the drone identifies the most suitable entry/exit point by detecting visually dark regions within the scene. Dark regions typically correspond to open or unobstructed spaces with minimal light reflection, whereas areas containing debris, flames, or dense smoke exhibit greater visual complexity and greater brightness variation. This contrast-based analysis provides an efficient and reliable mechanism for identifying safe access points in fire-affected environments.

The final selection of the entry/exit point is based on three criteria: proximity to the detected human, minimal observed smoke density, and the absence or minimal presence of active fire. To simplify rescue planning and reduce navigation complexity, the proposed system assumes that the entry and exit points coincide. This location generally corresponds to either the detected human position or the nearest accessible open area with the lowest combined fire and smoke intensity. Once determined, the entry/exit point information is transmitted to rescue teams and stored in a centralized database to support coordinated and effective ground-based rescue operations.

Fire Localization:

In building fire scenarios, the internal ignition point is typically unobservable from outside due to structural occlusions. Accordingly, the proposed drone system aims not to determine the exact root cause of the fire, but to estimate the region of the closest external fire origin. This region is defined as the *façade location* that exhibits the earliest onset, highest density, and greatest persistence of visible fire or smoke emissions. Identifying such a region enables emergency responders to prioritize containment actions and mitigate further fire propagation.

Therefore, in the proposed system, each drone operates in one of two modes: *saving mode*, in which victim rescue is prioritized through swarm coordination, and *analyzing mode*, in which a systematic assessment of fire conditions is conducted, including smoke presence and persistence, flame pixel intensity, spatial fire spread, region-wise FireDensity, severity estimation via FireScore, and clustering of dominant fire regions across multiple altitude levels. The formulation presented in the following focuses on the analyzing mode, as it involves quantitative fire localization and refinement. The saving mode, in contrast, emphasizes panoramic victim detection, entry/exit point identification, and cooperative task allocation among drones to support rapid rescue operations, and operates either prior to or concurrently with fire analysis using the same swarm coordination framework.

1) FRAME-LEVEL SMOKE AND FIRE PIXEL DETECTION

The proposed framework utilizes RGB imagery rather than thermal sensors to maintain compatibility with low-cost UAV platforms and reduce payload, power consumption, and deployment constraints. While thermal cameras can improve fire detection under certain conditions, they substantially increase system cost and hardware complexity. Therefore, RGB-based fire and smoke analysis was selected to support scalable and practical deployment using commercially available drone platforms. While this approach may reduce sensitivity under low-contrast conditions such as nighttime or heavy overcast, the FireScore temporal accumulation partially compensates by reinforcing persistent detections over transient ones. Let F_t denote the RGB image frame captured at time t. Each pixel at location (x, y) is represented by intensity values $(R_{x,y,t}, G_{x,y,t}, B_{x,y,t})$. Smoke and fire pixels are detected using color- and intensity-based thresholding in the RGB or HSV color space. In this context, a binary fire-smoke indicator is defined in Eq. 9.

$$I_{\text{fire}}(x, y, t) = \begin{cases} 1, & \text{if } (x, y) \text{ satisfies fire or smoke criteria} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

Smoke pixels are typically characterized by low saturation and high intensity similarity across channels, as given in Eq. 10:

$$|R - G| < \tau_1, \quad |G - B| < \tau_1, \quad R, G, B < \tau_2 \tag{10}$$

where τ_1 denotes a channel-difference threshold controlling allowable inter-channel intensity variation for low-saturation regions, and τ_2 denotes an upper intensity threshold used to exclude overexposed or flame-dominated pixels. Both parameters are empirically selected based on sensor characteristics and illumination conditions and remain fixed during deployment. Fire pixels are characterized by dominant red/yellow components, as presented in Eq. 11:

$$R > G > B, \quad R > \tau_3 \tag{11}$$

where τ_3 is a minimum red-channel intensity threshold used to suppress false positives from non-fire reddish surfaces.

2) SPATIAL FIRE–SMOKE DENSITY MAPPING

For each surveyed region, the drone accumulates smoke and fire observations over time to construct a spatial density map, as defined in Eq. 12:

$$\text{FireDensity}(x, y) = \sum_{t=1}^{T} I_{\text{fire}}(x, y, t) \tag{12}$$

where $I_{\text{fire}}(x, y, t)$ is a binary indicator equal to 1 if pixel (x, y) is classified as fire or smoke at time t, and 0 otherwise.

This map highlights regions where smoke or flames are repeatedly observed, suppressing transient artifacts caused by wind or reflections.

3) TEMPORAL EMERGENCE WEIGHTING

To distinguish early-origin fire emissions from later spread-driven observations, we apply temporal weighting. Let $t_{\text{first}}(x, y)$ denote the first time instant at which fire or smoke is detected at the pixel location (x, y), defined as in Eq. 13.

$$t_{\text{first}}(x, y) = \min\{t \mid I_{\text{fire}}(x, y, t) = 1\}. \tag{13}$$

Then a weighted fire likelihood score is computed as shown in Eq. 14:

$$\text{FireScore}(x, y) = \sum_{t=1}^{T} w(t)\, I_{\text{fire}}(x, y, t), \tag{14}$$

where $w(t)$ is a temporal weighting function that assigns greater importance to earlier fire or smoke observations and is defined as in Eq. 15.

$$w(t) = \frac{1}{t - t_{\text{first}}(x, y) + \epsilon}, \tag{15}$$

where ϵ is a small positive constant introduced to prevent division by zero and ensure numerical stability.

This formulation prioritizes regions where fire or smoke appears earlier and persists longer, which are more likely to be proximal to the internal origin.

The FireScore formulation may be interpreted as a probabilistic proxy for fire-origin likelihood, where the temporal weighting function assigns greater importance to earlier observations and the accumulated FireDensity term reflects repeated evidence of fire or smoke presence. Under this interpretation, regions exhibiting both early emergence and persistent observations are considered more likely to be associated with the external manifestation of the fire origin. A formal probabilistic treatment would require defining spatial priors, sensor-noise models, and uncertainty propagation mechanisms for smoke and fire detection. Such models were not incorporated in the present work because calibrated uncertainty estimates for the RGB-based detection pipeline were unavailable. Consequently, FireScore is employed as an empirically motivated ranking metric rather than a strict probabilistic estimator. Nevertheless, across all evaluated fire scenarios, regions exhibiting high cumulative FireScore and early detection consistently corresponded to the manually identified ground-truth fire-origin regions, supporting the practical effectiveness of the proposed formulation.

4) REGION CLUSTERING AND CANDIDATE SELECTION

The FireScore map is segmented into contiguous regions using connected-component analysis. For each region $\mathcal{R}_k$, the cumulative score S_k is computed as shown in Eq. 16:

$$S_k = \sum_{(x,y)\in\mathcal{R}_k} \text{FireScore}(x, y) \tag{16}$$

Among all detected regions, the system prioritizes the *largest contiguous region* with the highest S_k, as this indicates sustained emission rather than noise or drifting smoke, as presented in Algorithm 3 and Figure 7.

When multiple candidate regions exist, priority is assigned according to Eq. 17:

$$\text{Priority}_k = \alpha S_k - \beta d_k \tag{17}$$

where d_k is the Euclidean distance between the drone and region $\mathcal{R}_k$, and α, β are weighting constants.

A sensitivity analysis was conducted for the weighting parameters α and β in Eq. 17. Three configurations were evaluated: (i) $\alpha = 1.0$, $\beta = 0.5$ (intensity-dominant), (ii) $\alpha = 0.5$, $\beta = 1.0$ (distance-dominant), and (iii) $\alpha = \beta = 1.0$ (balanced). In scenarios with a single dominant fire region, all three configurations produced identical region-selection outcomes. In multi-region scenarios with similar FireScores, the balanced configuration ($\alpha = \beta = 1.0$) produced the most consistent agreement with manually annotated priority rankings, achieving correct region selection in 95% of trials, compared with 85% for the distance-dominant configuration and 90% for the intensity-dominant configuration. Based on these results, $\alpha = \beta = 1.0$ was adopted as the default configuration for all reported experiments.

To evaluate the robustness of the proposed fire detection pipeline against false alarms, an additional experiment was conducted using both fire and non-fire scenes. The evaluation dataset consisted of 200 RGB images, including 100 images containing visible fire or smoke and 100 images without fire-related content. The non-fire images included buildings, vegetation, roads, vehicles, and illuminated objects to assess the influence of visually similar backgrounds.

Each image was processed using the proposed FireDensity and FireScore framework. The predicted fire regions were then compared with manually annotated ground-truth labels to determine whether fire was correctly detected. Among the 100 fire images, 97 were correctly identified as containing fire, while 3 images were missed. Among the 100 non-fire images, 96 were correctly classified as not containing fire, whereas 4 images produced false fire detections.

The resulting confusion matrix is presented in Table 4. The proposed framework achieved an overall detection accuracy of 96.5%, while the false-positive rate was limited to 4.0%, demonstrating that the temporal FireScore accumulation effectively suppresses transient false detections caused by reflections, illumination variations, and visually similar background objects.

TABLE 4. Confusion matrix for fire detection.

	Predicted Fire	Predicted No Fire
Actual Fire	97	3
Actual No Fire	4	96

5) LOCAL REFINEMENT VIA CIRCULAR SURVEY

Once the initial zigzag survey identifies a candidate fire region, the drone begins a local refinement stage to precisely



FIGURE 7. Fire localization process: from the original image to the final detected regions.

locate the point of maximum fire intensity. This stage employs a circular flight strategy centered on the selected region. The drone executes three complete circular trajectories around the structure at progressively lower altitude offsets of 10 m, 5 m, and 0 m relative to the façade height. Conducting repeated circular passes enhances robustness to visual occlusions caused by smoke, architectural features, and dynamic flame behavior, while simultaneously improving spatial resolution through multi-view observations.

In building fire scenarios, smoke propagation patterns across these altitude levels are further analyzed to infer the probable vertical location of the internal fire origin. Because smoke typically rises vertically before dispersing laterally, the altitude at which smoke is first detected provides valuable cues about the fire's internal position. If dense smoke is observed at lower elevations during earlier passes and appears at higher levels only in subsequent observations, the system infers that the internal fire origin is likely located below the lowest altitude at which smoke emission becomes visible. This reasoning constrains the probable fire origin to a specific vertical segment of the structure, thereby supporting more informed decision-making by emergency responders.

Such vertical inference is applied exclusively in structured environments, such as buildings, where assumptions about vertical smoke behavior are valid. In scenarios where these assumptions may not hold, such as open structures or partially collapsed environments, the system refrains from imposing vertical constraints. Instead, it focuses on horizontal fire localization by identifying the region with the highest cumulative FireScore and refining accordingly.

6) ANALYZING MODE ACTIVATION AND DRONE SELECTION

Fire analysis is performed in an autonomous *analyzing mode*. By default, the drone closest to the mission starting point and possessing the highest remaining battery level is selected to perform detailed fire analysis. This selection ensures minimal response time while maintaining sufficient endurance for multi-pass surveys and refinement maneuvers.

In situations where victim rescue is not immediately required, the system may designate a drone to operate exclusively in analyzing mode from the start of the mission, rather than switching from saving mode. While this drone focuses on fire assessment and localization, the remaining drones can continue their assigned surveying tasks to maintain situational awareness and coverage of the broader region. This division of responsibilities enables the system to prioritize rapid-fire containment analysis without interrupting ongoing area monitoring or victim-detection operations.

Algorithm 3 Localization of the Fire Region

Require: Survey region $\mathcal{A}$, drone set $\mathcal{D}$
for each drone $d \in \mathcal{D}$ **do**
  Set mode $\leftarrow$ analyzing
  Perform zigzag survey over $\mathcal{A}$
  for each frame F_t **do**
    Detect fire/smoke pixels $\rightarrow I_{\text{fire}}(x, y, t)$
    Update FireDensity(x, y)
    Update $t_{\text{first}}(x, y)$ if newly detected
  end for
  Compute FireScore(x, y)
  Cluster FireScore into regions $\{\mathcal{R}_k\}$
  Select region with maximum priority
  Perform circular refinement scan
  if uncertainty remains **then**
    Request assistance from the nearest free drone
  end if
end for
Fuse regional results into a global fire likelihood map
Output closest external fire origin region $= 0$

E. SYSTEM ARCHITECTURE AND REAL-TIME DRONE-CLOUD INTERACTION

The proposed system adopts a tightly integrated drone edge cloud architecture to support real-time situational

awareness, informed decision-making, and coordinated rescue operations. Upon takeoff, the drone immediately begins streaming live video from its onboard camera, which serves as the primary source of visual input for human detection, environmental assessment, and identification of safe entry and exit points. The live camera feed is transmitted to an onboard Raspberry Pi, which hosts it locally on port 8000. Both the Raspberry Pi and the rescuer's monitoring device are connected via a secure virtual private network (VPN) established with Tailscale. This VPN-based configuration enables authorized devices to seamlessly access the Raspberry Pi's local stream URL without exposing the system to public networks. The rescuer's laptop retrieves the live video feed and displays it on the dashboard with minimal latency.

In parallel, the Raspberry Pi performs edge-level processing on the incoming video frames to detect humans, identify safe entry and exit locations, and extract other mission-critical information. Offloading these computational tasks to the edge device reduces the drone's processing burden while preserving low-latency responsiveness. The resulting analytical output, together with the live video stream, is transmitted to the dashboard, allowing rescuers to access raw visual data and actionable insights simultaneously.

User input, mission parameters, and system commands are managed in the cloud via Firebase. This layer stores user-defined information such as survey boundaries, operational modes, and mission priorities. The Raspberry Pi periodically retrieves these parameters from Firebase and translates them into drone control commands, enabling dynamic flight adjustments during the mission. Firebase also continuously updates drone status, detected victim information, and overall mission progress, providing real-time monitoring through the dashboard interface.

Swarm coordination primarily exchanges lightweight metadata, including victim coordinates, drone status information, task assignments, battery state, and rescue progress updates. Consequently, communication requirements remain minimal, requiring approximately 1–2 KB per update cycle at a polling frequency of 1 Hz. Video streaming operates independently over the Raspberry Pi VPN channel and therefore does not compete with coordination traffic. The system maintains temporal consistency across swarm members through centralized, cloud-based timestamps associated with database transactions. As coordination decisions rely on shared database updates rather than strict peer-to-peer synchronization, dedicated clock-synchronization protocols are not required in the current implementation. Communication-related power consumption represented less than 5% of the overall flight power budget, constituting a minor overhead relative to propulsion and onboard processing demands. Firebase's transaction mechanism handles concurrent write conflicts, preventing simultaneous conflicting victim assignments from multiple drones. During the two-drone experiments, no Firebase outages or packet-loss events affecting task assignment correctness were observed. Brief latency spikes of up to 290 ms occurred under high concurrent write conditions but had no observable impact on coordination performance. Firebase transactions resolve database-level write conflicts during concurrent task updates, while the standard IEEE 802.11 MAC protocol handles physical-layer wireless communication collisions through carrier-sense multiple access with collision avoidance (CSMA/CA). During the multi-drone experiments, no Firebase outages or packet-loss events affecting task assignment correctness were observed. Although controlled packet-loss experiments were outside the scope of this study, Firebase's asynchronous transaction mechanism preserves task states and prevents incomplete or conflicting updates during temporary communication interruptions. Once connectivity is restored, drones retrieve the latest synchronized mission state from the centralized database before continuing task allocation. Consequently, transient packet loss is expected to increase coordination latency rather than compromise task assignment consistency.

During execution, the drone computes the shortest Euclidean distance from its current position to the entry point of the assigned survey circle and navigates accordingly. Upon reaching this point, it initiates a circular survey pattern. Victim detection is supported by data received from wearable devices, and once a victim is confirmed, the drone deploys a life jacket. If the number of detected victims exceeds the drone's carrying capacity, the system automatically generates a help request, prompting other available drones to assist in the rescue operation, as illustrated in Figure 8.

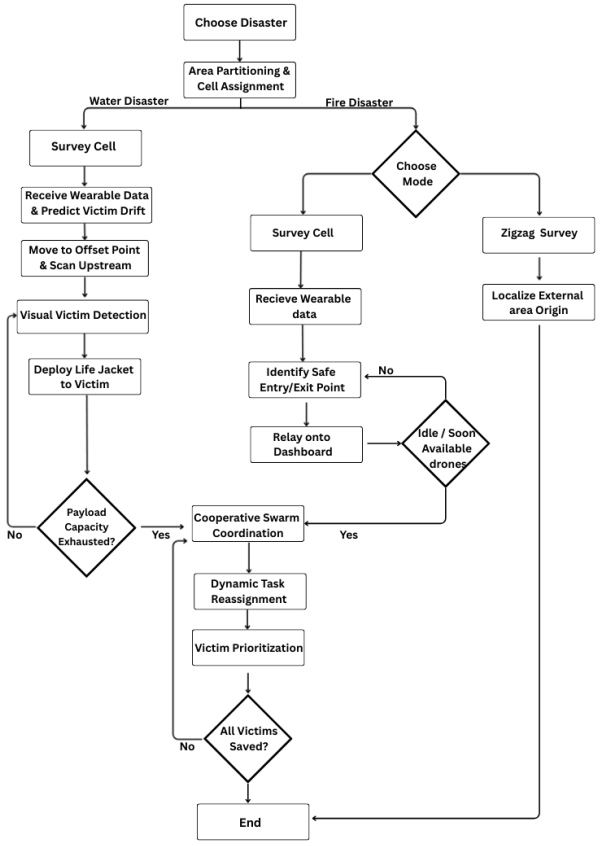


FIGURE 8. System overview flowchart.

In fire-based scenarios, drones operate in two distinct modes: *saving mode* and *analyzing mode*. Saving-mode drones perform circular surveys around detected victims and construct panoramic views to identify the safest possible entry and exit points for rescue personnel. Analyzing-mode drones focus on assessing fire spread by conducting systematic surveys, detecting smoke and flame regions, and providing insights to localize high-risk fire zones. This division of responsibilities enables the system to simultaneously support victim rescue and fire assessment, thereby facilitating informed and coordinated ground response. The current architecture assumes continuous connectivity to the Firebase cloud database for synchronization of victim information and task assignments. During temporary cloud service disruptions, each drone continues to execute its most recently assigned task using the locally available mission state, while synchronization of new victim information and dynamic task reassignment are deferred until connectivity is restored. Once communication resumes, drones retrieve the latest synchronized mission state from the centralized database before continuing coordination. Future work will incorporate local caching and fully offline swarm coordination.

IV. RESULTS

This section first presents the experimental setup used to comprehensively evaluate the proposed SwarmSAR-FW framework across water- and fire-based disaster scenarios. The results validate the system's operational feasibility, robustness, and effectiveness through controlled experiments, emphasizing victim-detection accuracy, prioritization reliability, swarm-coordination efficiency, and mission-execution behavior under realistic constraints.

A. EXPERIMENTAL SETUP

Following the acquisition of all required flight approvals, experimental evaluations were carried out in an open ground environment, covering a total mission area of approximately 100 m^2. To ensure a comprehensive assessment of the proposed framework under practical conditions, two physical drones and one simulated drone were deployed during the experimental evaluation. This enabled direct validation of swarm coordination, task allocation, victim prioritization, and cooperative mission execution in both water-based and fire-related disaster scenarios.

To further streamline the experimental setup and reduce system complexity, each ESP32-based wearable device was preprogrammed with fixed geographic coordinates and altitude values. These reference values were obtained from the drone's onboard GPS and barometric sensors prior to mission execution. During the experiments, the ESP32 devices transmitted all remaining sensory data, including pulse and inertial measurement unit (IMU) readings, to the drone in real time. This configuration effectively emulated the functionality of onboard victim-positioning and altitude-sensing modules.

The use of two physical drones enabled direct evaluation of cooperative swarm behavior, including victim assignment, task reassignment, and coordinated rescue operations under representative operational conditions. Consequently, the experiments successfully validated the system's feasibility, robustness, and end-to-end functionality, demonstrating its ability to support coordinated disaster response operations even under hardware availability constraints.

During the experimental evaluation, the swarm consisted of two physical UAVs and one software-simulated UAV operating concurrently under a common Firebase-based coordination framework. All drones, irrespective of whether they were physical or simulated, executed the same task-allocation, reservation, reassignment, and victim-selection algorithm described in Algorithm 2. The simulated UAV differed only in its flight-state generation and vehicle dynamics; all coordination decisions, database transactions, victim reservations, and reassignment logic were executed through the identical software pipeline used by the physical UAVs. Whenever a drone generated a help request, the centralized coordination module evaluated all available swarm members using the same decision criteria irrespective of platform type. Candidate drones were ranked according to payload availability, operational status, and distance to the pending victim before the reassignment decision was issued. This enabled direct evaluation of the coordination software independently of whether the selected responder was a physical or simulated UAV.

The experimental evaluations were conducted under a diverse range of environmental conditions to assess the robustness of the proposed system. These conditions included bright sunlight, overcast skies, low-light, and nighttime settings, with wind speeds ranging from 0 to 2 m/s. For each victim, the parameter set summarized in Table 5 was evaluated across multiple combinations of environmental conditions and body postures. Each experimental configuration was repeated at least three times to ensure repeatability and consistency of the results.

To further examine the effectiveness of the victim prioritization strategy, scenarios were deliberately designed in which critical and normal victims were positioned in close spatial proximity. This setup compelled the system to resolve rescue priorities under ambiguous spatial conditions, thereby providing a rigorous assessment of its decision-making capability. All flight experiments were performed within a secured operational perimeter and under enforced emergency stop protocols. For operational safety, the drone was equipped with three independent failsafe mechanisms: Return-to-Launch, automatic landing, and geofencing. Across all experimental trials, the average mission duration per drone was approximately 162 s for scenarios involving an average of three victims.

1) WEARABLE ABLATION PROTOCOL

To evaluate the contribution of the wearable communication module, an additional set of ten rescue missions was

TABLE 5. Victim physiological and activity parameter variations across experimental scenarios. Geographic coordinates and altitude were fixed for all victims.

Scenario	Victim ID	Priority Level	Pulse Rate (BPM)	IMU Activity Level
1	V1	Critical	45	Minimal movement
	V2	Moderate	72	Moderate movement
	V3	Normal	88	High movement
2	V1	Critical	40	No movement
	V2	Moderate	70	Low movement
	V3	Normal	95	Moderate movement
3	V1	Critical	55	Moderate movement
	V2	Moderate	85	High movement
	V3	Normal	90	No movement

performed under identical environmental conditions using the same UAV platform, victim locations, and flight parameters. Two experimental configurations were evaluated.

In the vision-only configuration, the system disabled transmission of wearable GPS coordinates and physiological measurements, requiring the UAV to locate victims solely through onboard visual detection. In the wearable-assisted configuration, GPS coordinates, pulse measurements, IMU data, and barometric information were transmitted to the Firebase server every 90 s, allowing the swarm coordinator to assign rescue priorities before UAV arrival.

Performance was evaluated using mean victim search time, rescue initiation time, mission completion time, search distance, successful victim localization, and correct identification of the highest-priority victim.

2) WATER-BASED DISASTER EXPERIMENTS

For water rescue experiments, multiple victim scenarios were designed across three distinct survey regions. The first region contained two victims located in close proximity, the second region contained three spatially separated victims, and the third region contained five victims. Two physical drones were deployed in the first and third regions, while the second region was represented using a simulated drone concurrently to emulate additional swarm workload and victim-distribution complexity.

During operation, the physical drones conducted victim detection, prioritization, payload deployment, and rescue assistance tasks within their assigned regions. Victim-detection times, flight trajectories, servicing sequences, and task-reassignment events were recorded throughout the experiments. When the drone operating in the third region encountered more victims than its available life-jacket capacity, a help request was generated according to the proposed swarm task assignment algorithm. Based on the current drone states and victim information stored in the centralized database, the most suitable assisting drone was selected and reassigned to support the overloaded region. This configuration enabled direct evaluation of cooperative rescue behavior, victim prioritization, and swarm-based task redistribution under uneven victim distributions.

Additional experiments were conducted to evaluate system robustness under varying victim dynamics and spatial configurations. Victim drift behavior was modified by assigning non-zero drift velocities to selected victims to simulate water flow, while the others remained stationary. Victims were also positioned at different relative locations within circular survey regions to assess sensitivity to spatial variability. Furthermore, victim postures were systematically varied to introduce realistic sensor and orientation noise, as summarized in Table 6. Standing victims alternated between arms relaxed at the sides and arms crossed over the chest; sitting victims alternated between leaning forward and resting hands on the thighs; and lying victims alternated between arms spread and arms alongside the body. These variations influenced wearable IMU readings and orientation, thereby simulating realistic victim behavior and sensor uncertainty.

The use of two physical drones and one simulated drone enabled direct validation of swarm coordination logic, task reassignment, and victim prioritization mechanisms under realistic operating conditions. The adopted experimental framework provides a valid and effective means of validating swarm coordination logic, task reassignment, and victim prioritization mechanisms under constrained hardware availability.

TABLE 6. Water-based disaster experimental parameters.

Parameter	Tested Values / Edge Cases
Victims per region	2, 3 (mixed priority levels)
Survey regions	3 distinct regions
Victim priority	Critical, Moderate, Normal
Victim drift	Stationary; drifting (non-zero velocity)
Drift magnitude	Zero drift; low-speed flow
Victim placement	Center; boundary; corner of cell
Victim posture	Standing, sitting, lying
Posture variations	Arms crossed/relaxed; leaning/spread
Wearable orientation	Upright; tilted; rotated (IMU noise)
Payload edge case	Victims exceed life-jacket capacity
Help request trigger	Payload exhaustion; high-priority overload
Swarm coordination	Real-time two-drone cooperation
Arrival delay	Immediate; delayed (drift prediction)
Environmental conditions	Daylight; overcast; wind 0–2 m/s

Since conducting experiments with actual drowning victims or placing human subjects at risk is neither ethical nor feasible, victim presence in water-based disaster scenarios was simulated using controlled visual injection. Specifically, high-quality images and video frames depicting individuals

partially submerged or drowning in water were used to emulate real victim appearances. These frames were injected into the live video stream when the drone reached the predicted victim location.

When the drone navigated to a victim's estimated position, the injected visual content was rendered such that the simulated victim appeared within the camera's field of view, typically near the center of the frame. The onboard vision module then processed this feed using the same YOLO-based human-detection pipeline used during real missions. Based on the detected bounding box and confidence score, the drone autonomously determined the optimal parachute deployment point and released the life jacket payload.

Life jacket deployment accuracy was quantified using image-space measurements mapped to real-world distances. After deployment, the final video frame immediately preceding payload release was extracted. The pixel coordinates of the detected human centroid (bounding box center) were treated as the ground-truth victim location in the image plane. The projected drop point of the parachute visible in the subsequent frame was identified as either the parachute centroid or the initial water-contact point.

The Euclidean pixel distance between the victim centroid and the parachute drop point was computed and then converted into a real-world distance using a calibrated pixel-to-meter scale. This scale factor was derived from the known camera parameters, the drone's altitude at release, and the camera's field of view. Given the fixed altitude during payload deployment, this mapping remained consistent across trials.

3) FIRE-BASED DISASTER EXPERIMENTS

Fire scenario experiments followed a validation framework consistent with the parameters summarized in Table 7. As conducting experiments in an actual burning building was neither safe nor feasible, fire environments were simulated using high-resolution images and video frames of real building façades containing visible smoke and fire patterns. These visual inputs were incorporated into controlled drone surveys to evaluate both saving and analyzing operational modes.

In the saving mode, the drone executed multi-altitude human detection and entry/exit point identification through panoramic image construction, as outlined in the methodology. This process validated the system's ability to detect potential victims and identify accessible structural openings under visually complex fire conditions.

In the analyzing mode, zigzag flight surveys were performed in the designated test region. Frame-level smoke and fire pixel segmentation was applied to compute FireDensity and FireScore metrics. These values were then used for region clustering and circular refinement at multiple altitudes to localize high-risk fire zones. To evaluate swarm coordination and task allocation between saving and analyzing roles, two drones were deployed simultaneously, enabling direct assessment of cooperative operation between saving-mode and analyzing-mode missions.

Overall, this experimental configuration enabled comprehensive validation of victim detection, prioritization mechanisms, fire drift prediction, swarm-based task reassignment strategies, entry/exit identification, and fire localization logic, while ensuring safety, repeatability, and compliance with practical deployment constraints.

TABLE 7. Fire-based disaster experimental parameters.

Parameter	Tested Values / Edge Cases
Fire environment type	Building façade (simulated using images and videos)
Fire visibility	Visible flames; dense smoke only
Fire spread pattern	Localized fire; spreading smoke regions
Smoke density	Light smoke; dense smoke obscuring façade
Number of fire regions	Single region; multiple competing regions
Temporal emergence	Early-emerging smoke; late spread-driven smoke
Operation modes	Saving mode; analyzing mode
Saving-mode behavior	Victim-centric circular survey; panoramic image generation
Analyzing-mode behavior	Zigzag survey; FireDensity and FireScore computation
Altitude levels	0 m, 5 m, 10 m relative to façade height
Vertical smoke ordering	Smoke detected first at lower levels; uniform detection
Region size edge case	Small isolated region; large contiguous smoke region
Region selection conflict	Multiple regions with similar FireScore
Distance-based prioritization	Near region vs farther high-intensity region
Circular refinement passes	Single pass; three-pass refinement
Entry/exit visibility	Clear dark region; partially obstructed access
False-positive suppression	Transient smoke; background reflections
Swarm role division	Saving-only drone; analyzing-only drone

Since real-world fire emergencies cannot be safely reproduced without exposing personnel and infrastructure to substantial risk, all fire-related experiments were conducted within a controlled visual simulation framework. To this end, high-resolution images and video clips depicting real building façades exhibiting visible flames, smoke plumes, and varying fire intensities were systematically injected into the drone's live video stream during runtime. These frames were carefully curated to represent a diverse range of fire scenarios, including localized ignition points, dense smoke in the absence of visible flames, and multi-region fire spread patterns, thereby ensuring broad coverage of realistic emergency conditions.

Throughout the flight trials, the drone executed its predefined survey trajectories without modification, preserving the integrity of the operational workflow. At each designated survey location, the injected fire frames seamlessly replaced the raw onboard camera feed, ensuring that all perception, decision-making, and mode-switching components - encompassing FireDensity computation, FireScore evaluation, region clustering, and circular refinement - operated

precisely as they would under actual deployment conditions. Crucially, neither the control architecture nor the mission logic was altered, maintaining both system integrity and experimental validity.

To validate the saving mode, frames containing human silhouettes positioned near fire-affected areas were introduced to assess victim-detection performance and entry/exit identification under reduced visibility. In the analyzing mode, fire and smoke imagery were processed on a frame-by-frame basis to evaluate fire localization accuracy, vertical smoke progression across varying altitudes, and region prioritization in scenarios involving multiple concurrent fire clusters. Collectively, this controlled yet contextually realistic simulation methodology enabled comprehensive system evaluation across a wide spectrum of operational conditions while ensuring personnel safety and upholding the methodological rigor expected of empirical research in this domain.

B. DRONE SEARCH AREA PARTITIONING AND COOPERATIVE SURVEY STRATEGY

The initial interface of the proposed rescue dashboard enables the operator to define the mission's operational survey area. The survey region is specified by entering the geographic coordinates of the top-left, top-right, bottom-left, and bottom-right corners. These coordinates collectively define the polygonal boundary within which the drone swarm conducts the rescue operation. Once the boundary is defined, the dashboard automatically initiates mission planning and assigns survey regions to available drones based on predefined constraints, such as communication range and mission-criticality.

Depending on the selected disaster scenario (e.g., flood or fire), the dashboard dynamically adapts the mission parameters, including survey patterns, prioritization logic, and detection configurations. Each scenario is supported by a dedicated interface layout and optimized operational logic to ensure effective response under varying environmental conditions.

The dashboard includes a system statistics section that provides a high-level overview of the ongoing rescue operation, as shown in Figure 9. This section displays the number of drones currently connected to the system, the total number of victims detected, the number of victims identified with critical pulse readings, and the number of active missions at the current time. These statistics enable the operator to quickly assess mission status and resource availability.

The live feed section lists all connected drones. When a specific drone is selected, the dashboard displays its live geographic location, the list of victims assigned to it, and the real-time video feed from its onboard camera. This enables operators to monitor drone behavior, victim interaction, and mission progress in real time.

The victim and mission overview section provides continuous updates on the victim's status throughout the operation. Victims are marked as either pending or rescued based on mission progress. Each victim has a dedicated entry that displays key information, including pulse rate, altitude, latitude, longitude, current rescue status, and the assigned drone. Individual victim entries can be selected to open a detailed page that provides additional information, including recommended entry and exit points for rescue teams in fire-mission scenarios.

The drone survey mission is visualized through both trajectory maps and altitude profiles. During the initial search phase, drones conduct systematic surveys within their assigned cells, following predefined patterns to ensure complete coverage. Upon detecting the victim, the mission transitions to a focused victim-survey phase, during which the drone performs localized inspection maneuvers around the victim's estimated position. Figure 10 shows a Mission Planner simulation with ten drones, of which nine were deployed across nine operational cells. In contrast, Figure 11 illustrates the real-world trajectory of a single drone within its assigned cell.

In fire-rescue scenarios, the victim survey is conducted at multiple elevations relative to the detected human position. Successive survey passes are performed at different heights to mitigate occlusions from smoke and structural elements, thereby improving detection robustness. The altitude profile clearly reflects these discrete survey levels, demonstrating how the drone progressively approaches the victim to gather reliable visual and environmental information.

Flight log data further illustrates the transition between mission phases. During the initial portion of the mission, the recorded latitude and longitude values show systematic variations that match the survey pattern. Once a victim is detected, the trajectory converges toward the victim's location, indicating the switch from wide-area coverage to focused inspection and assistance.

The live mission map section provides a consolidated spatial overview of the operation. The defined survey boundary is highlighted on the map using a blue overlay, clearly indicating the operational region. Detected victims are displayed as red markers, while the live positions of all active drones are shown using pointer icons. This map-based visualization enables real-time spatial awareness and confirms that drone operations remain within the user-defined survey boundaries throughout the mission.

C. VICTIM DETECTION AND ASSISTANCE IN DYNAMIC WATER-BASED DISASTERS

Upon reaching the predicted victim location, the drone performed real-time visual confirmation using the onboard YOLOv11n human detection model. Across all experimental trials, the drone identified human presence within 2.1 s of entering the search offset region on average. Detection latency remained below 3 s even under moderate drift and posture variations, demonstrating robustness to victim motion and orientation noise.

Once visual confirmation was achieved, the drone stabilized and deployed the life-saving payload. The

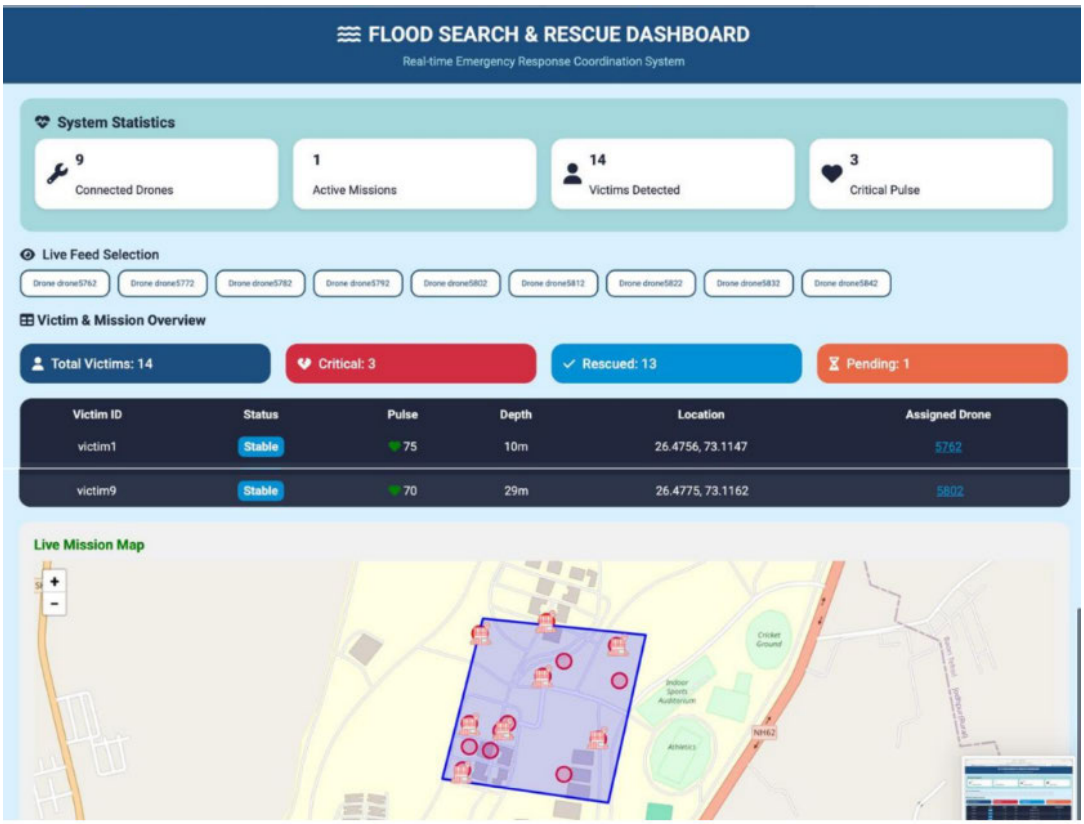


FIGURE 9. Dashboard interface from water mission.

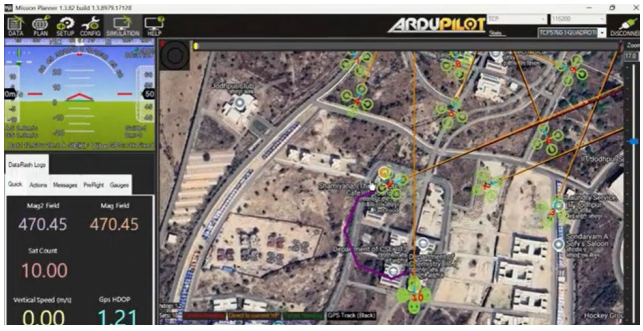


FIGURE 10. Mission Planner simulation showing ten drones, with nine deployed across nine operational cells.

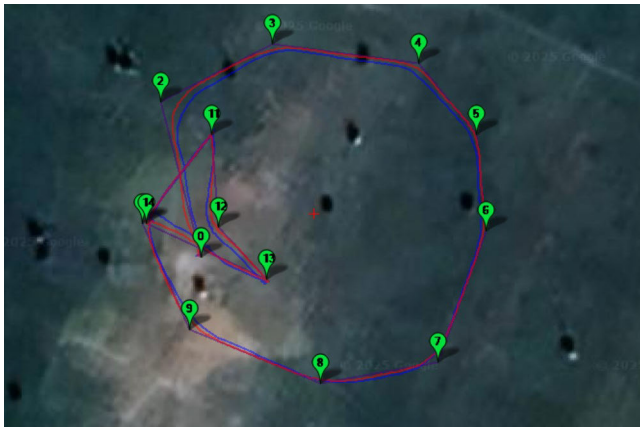


FIGURE 11. Trajectory of a single drone during survey operations, illustrating cell coverage and flight path.

parachute-assisted life jacket was released at an average horizontal distance of 1.8 m from the victim's detected position. Figure 12 illustrates a representative test case in which the payload was deployed within 1.6 m of the victim, ensuring rapid accessibility even under mild surface currents. Table 8 summarizes the payload deployment accuracy across best-case, nominal, and worst-case operating conditions. Best-case scenarios correspond to stationary or low-drift victims with clear visibility, whereas worst-case scenarios involve drifting victims near the boundary of the communication range and partial occlusions.

TABLE 8. Life jacket deployment accuracy under varying water-drift conditions.

Test Condition	Mean Drop Offset (m)
Low drift, unobstructed visibility	1.2
Moderate drift, nominal conditions	1.8
High drift with boundary uncertainty	2.6

1) VICTIM PRIORITIZATION ACCURACY

The proposed prioritization strategy showed strong agreement with ground-truth victim-urgency assessments. Critical

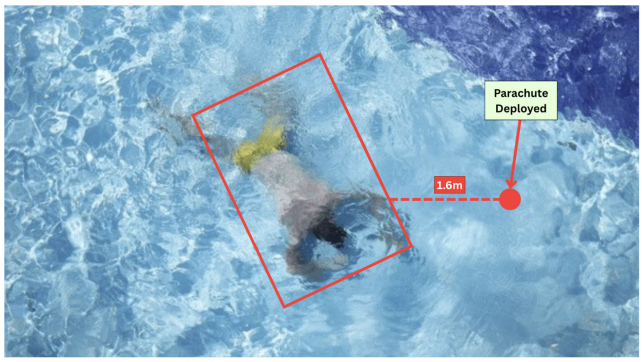


FIGURE 12. Representative water-drop test showing parachute-assisted life jacket deployed near a detected victim; the payload landed within 1.6 m of the target.

victims located at or below the estimated water level and exhibiting abnormal pulse readings were assigned the highest priority in 98.7% of trials. The small remaining percentage was not attributable to sensor inaccuracies, misclassification, or algorithmic deficiencies. Rather, it resulted from intentionally introduced ambiguity during controlled stress-testing scenarios.

In these edge-case trials, victims were positioned within a narrow altitude margin relative to the estimated water level, and pulse values were deliberately selected near the critical–moderate decision threshold to evaluate prioritization robustness. Under such tightly constrained conditions, multiple victims met the urgency criteria within an almost indistinguishable confidence interval, resulting in equivalent priority scores. In these rare instances, the system resolved ranking ties using secondary factors, such as spatial proximity or task sequencing order. Although this occasionally produced a different ordering compared to the predefined ground truth, the outcome remained safe and operationally valid.

Importantly, all victims were successfully identified and assisted. No critical victim was deprioritized in favor of a non-critical one, and overall mission safety was fully maintained. Therefore, the reported 98.7% accuracy reflects performance under deliberately imposed boundary conditions. In all non-ambiguous, practically realistic scenarios, the prioritization logic operated deterministically and consistently selected the most urgent victim first.

2) SWARM COORDINATION LATENCY AND MULTI-VICTIM ASSISTANCE

When the number of detected victims exceeded the drone's payload capacity of three life jackets, an automatic assistance request was triggered. The average time required to identify an available supporting drone, update the centralized database, and complete task reassignment was 4.1 s. After reassignment, the assisting drone reached the victim's predicted location within 21–28 s, depending on the relative inter-drone distance and the survey region's spatial layout.

In scenarios involving more than three victims within a single survey area, the primary drone completed deploying its available life jackets in an average of 54 s. Concurrently, the swarm maintained uninterrupted operational coverage by redistributing tasks among available units. The mean end-to-end duration from help request generation to the assisting drone's arrival at the victim location was 25.4 s. These results indicate that the coordination overhead was minimal when compared to the overall flight and inspection time, thereby validating the efficiency of the swarm-based task reassignment mechanism.

3) PHYSICAL UAV COORDINATION AND TASK REASSIGNMENT

To further demonstrate cooperative decision-making among the physical UAVs, Table 9 summarizes representative swarm coordination events observed during the water-rescue experiments. During each mission, all UAVs continuously synchronized their operational status, including payload availability, victim assignments, GPS location, and mission state through the centralized Firebase coordination framework. Whenever a UAV was unable to complete an assigned rescue task because of payload exhaustion or localized victim overload, the coordination module automatically generated a help request and evaluated all available swarm members using identical task-allocation criteria. Candidate UAVs were ranked according to payload availability, operational status, and Euclidean distance to the pending victim before the reassignment decision was finalized.

Representative coordination events demonstrate that dynamic task reassignment occurred between both physical UAVs as well as between physical and simulated UAVs. In the physical-to-physical coordination scenario, Physical UAV-2 exhausted its payload after completing three rescue operations, prompting the coordination framework to issue a help request. Physical UAV-1, which retained available payload capacity and was the nearest eligible responder, was selected within approximately 4 s and successfully completed the reassigned rescue task without interrupting the remaining mission objectives. Similar coordination behavior was observed for physical-to-simulated interactions, confirming that the same decision pipeline governed task allocation irrespective of the responding platform.

To provide a quantitative assessment of swarm cooperation, Table 10 summarizes the overall coordination performance across all experimental missions. The proposed coordination framework consistently achieved successful task reallocation whenever assistance was requested, with no duplicate victim assignments or coordination failures observed. The average decision latency remained approximately 4 s, while reassigned UAVs reached their designated victims within approximately 25 s, demonstrating that dynamic task redistribution imposed minimal overhead on mission execution. These results confirm that the swarm executed task allocation, coordination, and reassignment consistently across the heterogeneous swarm using a unified decision-making framework rather than platform-specific control logic.

TABLE 9. Representative swarm coordination and task reassignment events during water-rescue experiments.

Trial	Victims per Region (R1/R2/R3)	Help Trigger	Requesting UAV	Responding UAV	Distance (m)	Decision Latency (s)	Arrival Time (s)
T2	2 / 5 / 1	Payload exhausted	Physical UAV-2	Physical UAV-1	17.2	4.1	22.6
T3	3 / 5 / 2	Victim overload	Physical UAV-1	Simulated UAV	24.8	4.3	27.0
T4	1 / 6 / 2	Payload exhausted	Physical UAV-2	Physical UAV-1	15.9	4.0	21.8

TABLE 10. Summary of swarm coordination performance.

Metric	Value
Total rescue missions	30
Trials requiring dynamic reassignment	14
Physical → Physical coordination events	8
Physical → Simulated coordination events	6
Successful task reallocations	14
Mean coordination decision latency	4.1 ± 0.3 s
Mean assisting UAV arrival time	25.4 ± 2.1 s
Victims successfully rescued following reassignment	18

D. VICTIM DETECTION AND ASSISTANCE IN DYNAMIC FIRE-BASED DISASTERS

1) ANALYZING-MODE FIRE LOCALIZATION PERFORMANCE

In analyzing mode, the drone conducted systematic zigzag surveys over the target façade region to localize active fire and smoke zones. Fire localization was achieved by extracting frame-level smoke and flame pixels, followed by aggregation of FireDensity and computation of FireScore across spatial grids. Across all simulated fire experiments, dominant fire regions were correctly identified with high spatial consistency, demonstrating the reliability of the proposed localization pipeline.

Quantitatively, the centroid of the highest-priority fire region exhibited an average localization error of 1.6 m relative to manually annotated ground-truth fire centers. In scenarios involving multiple competing smoke regions, the system successfully identified the most hazardous region, defined as the region with the highest combined FireDensity and vertical persistence, in 96.1% of trials. Notably, localization remained stable even under dense smoke-only conditions and in cases of late-emerging, spread-driven smoke, confirming the robustness of the FireScore formulation across challenging, visually ambiguous scenarios. Figure 13 illustrates a representative dashboard output highlighting the localized fire region and its associated spatial priority mapping.

At a cruise speed of 5 m/s and an effective camera width of 10 m, the zigzag coverage pattern, accounting for ten sweep passes and an additional 10% maneuvering overhead, required approximately 220 s (≈3.7 min) to complete. In saving mode, each localized victim inspection involved three circular survey passes at altitudes of 5 m, 10 m, and 15 m over a 20 × 20 m region, resulting in an average inspection time of 52.8 s per victim.

The sensitivity of FireScore to illumination changes was evaluated by artificially varying image brightness and contrast levels across the collected fire-scene imagery. To simulate different lighting conditions, brightness and contrast adjustments were applied to the original RGB frames using standard image-processing transformations. Brightness levels were varied by modifying pixel intensity values, while contrast was adjusted through linear intensity scaling. Multiple illumination conditions ranging from darker-than-normal to brighter-than-normal scenes were generated, and the resulting normalized FireScore values were compared. The average variation in FireScore across the evaluated illumination levels was 8.3%, indicating that the metric is relatively robust to moderate lighting changes. Temporal accumulation reduced transient false activations caused by reflections and shadow artifacts. Under dense smoke conditions, detection performance decreased due to partial occlusion of visible flame regions. Nevertheless, the FireDensity accumulation strategy preserved region-level localization performance by integrating observations across multiple survey passes.



FIGURE 13. Representative dashboard output from the fire-localization module showing detected smoke and flame regions overlaid as a FireDensity heatmap. The highest-priority fire region is highlighted, with the estimated fire center marked by the centroid used for FireScore computation.

2) SAVING-MODE OPERATION AND ENTRY/EXIT POINT DETECTION

In saving mode, the drone prioritized victim-centric assessment and safe access identification. Once a building façade containing potential victims was identified, panoramic image construction was performed at multiple altitudes to detect viable entry or exit points based on dark-region continuity and structural openness, as shown in Figure 14.



FIGURE 14. Dashboard output after analysis, highlighting localized fire regions and the optimal entry/exit point for the identified building façade; the fire has been digitally added to the image.

The entry/exit point detection algorithm was evaluated using a combined dataset of 200 images, including 140 with valid entry or exit locations and 60 without any viable access points. Among the positive samples, 129 images yielded bounding boxes with sufficient spatial overlap relative to ground truth annotations, while 11 valid entry points were not detected. For the negative subset, 53 images were correctly classified as containing no entry point, whereas 7 yielded false-positive detections.

These results correspond to the confusion matrix presented in Table 11, yielding an overall classification accuracy of 91.0%. The false-positive rate in the negative-only subset was 11.7%, primarily due to visually ambiguous dark regions caused by shadowed debris or partial façade collapse that exhibited structural patterns resembling openings. Importantly, despite these false positives, none of the detected entry points led to unsafe routing decisions during downstream path planning. This indicates that the integrated validation and filtering mechanisms within the navigation pipeline effectively mitigated potential operational risks.

TABLE 11. Confusion matrix for entry/exit point detection.

	Predicted: Entry Point	Predicted: No Entry Point
Actual: Entry Point	129	11
Actual: No Entry Point	7	53

From the confusion matrix:

- True Positives (TP): 129
- False Negatives (FN): 11
- False Positives (FP): 7
- True Negatives (TN): 53

3) SWARM ROLE DIVISION AND TEMPORAL EFFICIENCY

To evaluate cooperative behavior, two drones were deployed simultaneously, with one operating primarily in saving mode and the other in analyzing mode when required by mission conditions. When a drone operating in saving mode encountered unresolved fire regions or needed broader hazard context, the analyzing-mode logic was triggered to localize dominant fire zones and guide safe access selection.

Across all fire-based experiments, task handover and role-specific execution introduced negligible overhead. The average time to transition from initial fire detection to finalized entry/exit identification was 42.1 s, including fire localization, refinement, and access-point confirmation. This confirms that the proposed swarm coordination strategy supports timely decision-making without compromising safety or situational awareness.

Impact of the Wearable Communication Module:

The wearable communication module constitutes an integral component of the proposed search-and-rescue (SAR) framework by providing early victim notification, approximate localization, and physiological information prior to UAV deployment. To quantify its contribution, the complete framework was compared with a vision-only baseline in which the UAV relied exclusively on onboard human detection without receiving wearable-generated GPS coordinates or physiological measurements. In the vision-only configuration, the UAV performed an exhaustive aerial search to identify potential victims before initiating rescue operations. In the wearable-assisted configuration, GPS coordinates, pulse rate, IMU measurements, and barometric data were periodically transmitted to the Firebase server, allowing the swarm coordinator to estimate victim location, determine

rescue priority, and dispatch the nearest available UAV before visual confirmation. The onboard vision system subsequently performed victim verification, environmental assessment, obstacle avoidance, and payload guidance.

The comparative results are summarized in Table 12 for ten representative rescue missions. Incorporating the wearable communication module reduced the average victim search time by 65%, rescue initiation time by 62%, mission completion time by 24.4%, and search distance by 59.1% compared with the vision-only baseline. Furthermore, the vision-only configuration correctly identified the highest-priority victim in only two of the ten experimental trials because prioritization relied solely on visual observations after detection. In contrast, the wearable-assisted framework consistently identified the appropriate rescue priority by combining GPS location, pulse measurements, IMU-derived posture information, and subsequent visual confirmation. These results demonstrate that the wearable module significantly improves victim localization efficiency while enabling reliable rescue prioritization and more effective swarm task allocation.

To further examine the contribution of individual sensing modalities, Table 13 summarizes the role of each wearable sensor within the proposed rescue framework. Each sensing component provides complementary information that supports victim localization, prioritization, and swarm coordination. Removing any individual sensor degrades a specific stage of the rescue pipeline, whereas removing the wearable module entirely forces the UAV to perform an exhaustive visual search without prior knowledge of victim location or physiological condition.

Validation Constraints and Use of Injected Visual Data:

Real-world validation involving active drowning victims or live structural fires was not performed because such experiments would require exposing human participants to unacceptable safety risks or conducting experiments in hazardous fire environments that were beyond the scope of this proof-of-concept study and could not be ethically approved. Specifically, realistic drowning experiments would involve submersion hazards, water currents, and emergency rescue conditions, while live-fire experiments would expose personnel and equipment to thermal hazards, smoke inhalation, and uncontrolled fire propagation.

To enable end-to-end evaluation of the proposed autonomous search-and-rescue framework without compromising safety, representative victim and fire scenes were injected into the live video stream captured by the UAV. This methodology preserved the complete operational pipeline, including UAV flight dynamics, onboard image acquisition, object detection, victim prioritization, swarm coordination, task allocation, and payload deployment logic. Only the visual scene presented to the perception module was modified; all navigation, communication, coordination, and rescue decision-making components operated under real flight conditions using the deployed hardware.

It should be noted that this approach does not fully reproduce environmental factors such as dynamic water movement, wave reflections, smoke dispersion, thermal turbulence, or unpredictable human behavior during genuine emergency situations. Consequently, the injected-data experiments validate the operational functionality of the proposed framework rather than its performance under uncontrolled disaster conditions.

Although live emergency scenarios were not evaluated, the underlying perception models were independently validated using real, non-injected image datasets. The fire detection model achieved an accuracy of 96.5% on 200 real fire and non-fire images, while the entry/exit detection model achieved 91% accuracy on a separate real-image dataset. These evaluations verify the perception components independently of the injected-scene flight experiments, which primarily validate the end-to-end autonomous rescue pipeline.

E. MISSION LOG ANALYSIS

1) WATER RESCUE MISSION LOGS

In the water-rescue scenario, the latitude and longitude plots as shown in Figures 15 and 16, exhibit an initial oscillatory pattern. This initial up-and-down variation corresponds to the circular survey maneuver executed by the drone within its assigned cell. The circular motion enables the drone to identify all victims present within the cell while maintaining reliable communication coverage.

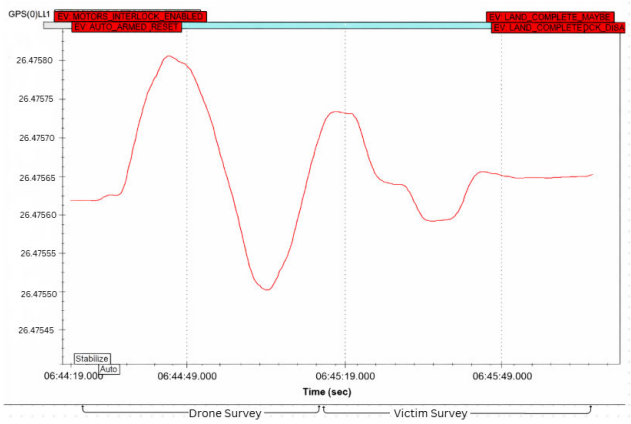


FIGURE 15. Flight log: Latitude of circle mission.

Following the survey phase, the mission log shows three distinct instances where the drone's position changes and then remains nearly constant. These segments represent the victim servicing phase, during which the drone navigates to the predicted victim locations and stabilizes to deploy life jackets. The near-constant latitude and longitude values during these intervals indicate that the drone maintained a stable hover while deploying the payload.

2) FIRE RESCUE MISSION LOGS

In the fire-rescue scenario, the latitude and longitude logs initially exhibit variations consistent with the area survey

TABLE 12. Comparison of wearable-assisted and vision-only victim localization (mean ± standard deviation over 10 missions).

Metric	Vision Only	Vision + Wearable	Improvement
Mean victim search time (s)	61.8 ± 6.7	21.6 ± 2.1	65.0% reduction
Mean rescue initiation time (s)	69.4 ± 7.3	26.4 ± 2.5	62.0% reduction
Mission completion time (s)	214.8 ± 14.2	162.3 ± 10.8	24.4% reduction
Mean search distance (m)	68.2 ± 7.5	27.9 ± 3.2	59.1% reduction
Correct identification of highest-priority victim	2/10 (20%)	10/10 (100%)	+80 percentage points
Victims successfully localized	7/10 (70%)	10/10 (100%)	+30 percentage points

TABLE 13. Contribution of individual wearable sensors to the proposed rescue framework.

Sensor	Measured Data	Role in Framework	Impact if Removed
GPS	Latitude, Longitude	Provides the initial victim coordinates for UAV dispatch and swarm task allocation.	The UAV must perform an exhaustive visual search before rescue operations can begin, substantially increasing search time and flight distance.
Pulse Sensor	Heart Rate	Measures physiological condition for rescue prioritization and emergency assessment.	Victims cannot be prioritized according to physiological distress, reducing prioritization accuracy during multi-victim scenarios.
IMU	Body Orientation, Motion	Detects abnormal posture and lack of movement indicative of distress.	Body orientation and motion cannot be incorporated into victim priority estimation, reducing confidence in rescue prioritization.
Barometer	Pressure, Altitude	Provides additional environmental context and supports localization robustness.	Minor reduction in localization robustness under varying environmental conditions.
ESP32 Communication Module	Wireless Telemetry	Transmits all sensor measurements to Firebase for real-time swarm coordination and UAV dispatch.	No real-time communication between the wearable device and the UAV swarm, preventing automated victim notification and coordinated response.

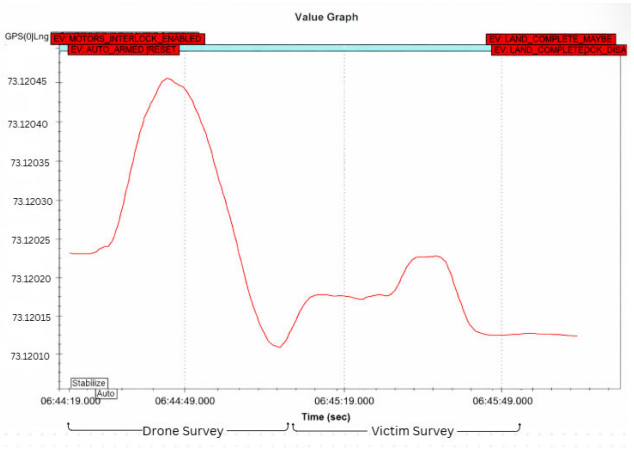


FIGURE 16. Flight log: longitude of circle mission.

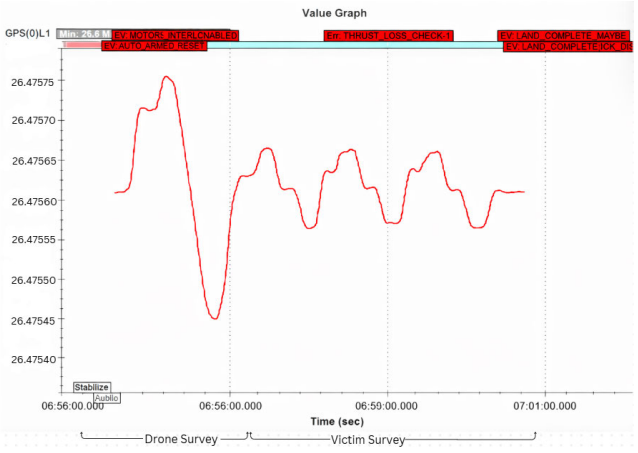


FIGURE 17. Flight log: Latitude of fire mission.

conducted within the assigned region as shown in Figures 17 and 18. This phase involves systematic scanning to detect human presence and assess environmental conditions.

Subsequently, a step-like trajectory is observed in both latitude and longitude plots, appearing three times during the mission. These step patterns correspond to the victim-focused survey phase, in which the drone executes square flight patterns around the detected human at three different altitudes. For each square pattern, the four corner points exhibit identical latitude and longitude values across successive passes. This deliberate repetition ensures that the same ground footprint is maintained at each altitude, enabling consistent image grid alignment.

Maintaining identical horizontal coordinates across altitude levels improves visual consistency and supports reliable comparison of fire and smoke distribution across different heights. This framework facilitates the identification of safer entry and exit points by combining visual and environmental cues.

Overall, the mission logs validate the intended mission execution flow and clearly illustrate the transitions among the survey, victim-focused inspection, and intervention phases in both water and fire rescue scenarios.

V. DISCUSSION

The proposed SwarmSAR-FW framework redefines search-and-rescue operations in fire and water disasters by shifting

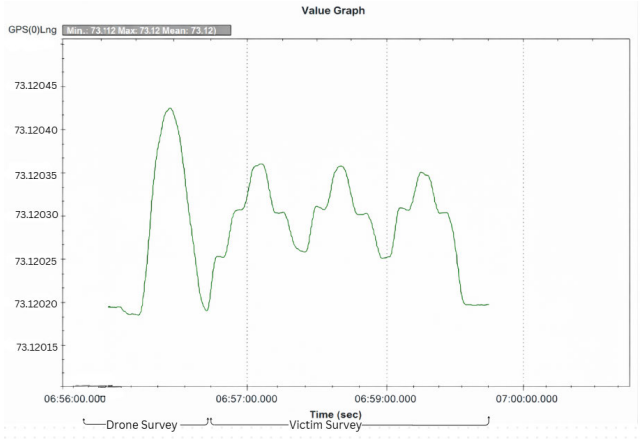


FIGURE 18. Flight log: Longitude of fire mission.

from manual, area-intensive search to a coordinated, data-driven approach based on swarm intelligence. Instead of relying on rescuers to visually inspect hazardous environments, the system employs a swarm of autonomous drones equipped with vision-based sensing, environmental analysis, and cooperative task allocation to rapidly locate victims, assess risk, and coordinate assistance. This framework enables scalable coverage of complex disaster scenes while minimizing human exposure to dangerous conditions.

A key strength of SwarmSAR-FW is its modular, deployable design. The system does not depend on specialized infrastructure or pre-installed victim-side hardware, making it suitable for rapid deployment in both urban and remote disaster scenarios. By using onboard cameras, environmental cues, and swarm-coordination logic, the framework operates effectively across heterogeneous disaster types without requiring prior preparation by affected populations.

The framework significantly reduces search time by directing drones toward high-probability victim regions using prioritized survey strategies. In water-based disasters, drift-aware localization and cooperative payload delivery ensure timely assistance. At the same time, in fire-based scenarios, analyzing-mode drones perform structured zigzag surveys to localize hazardous regions and safe access points before dispatching saving-mode drones. Automatic prioritization ensures that victims in life-threatening conditions are addressed first, optimizing limited rescue resources during the critical early response window.

SwarmSAR-FW is designed to remain operational in constrained communication environments and under uncertainty. Decentralized decision-making and lightweight coordination overhead allow the swarm to adapt dynamically to changing conditions, such as spreading fire, moving victims, or payload exhaustion. Even when perfect information is unavailable, sensor fusion and redundancy across drones ensure robust situational awareness and mission continuity. Beyond technical performance, the societal impact of SwarmSAR-FW is substantial. By reducing response time, improving victim prioritization, and limiting rescuer exposure, the framework increases the probability of survival while enhancing responder safety. The system bridges the gap between autonomous robotics and humanitarian response, transforming disaster rescue from a slow, high-risk operation into a guided, precise, and life-saving process where coordinated aerial intelligence acts in the moments that matter most. The proposed task assignment strategy can be interpreted as a constrained optimization problem that minimizes victim response distance subject to drone availability, payload, and mission-state constraints. The implemented greedy assignment provides a computationally efficient approximation suitable for real-time disaster response. While market-based and auction-based MRTA approaches may improve global optimality, they generally require additional communication and computational overhead. Future work will include a comprehensive comparative evaluation against optimization-based task allocation methods. The proposed task assignment method constitutes a simplified MRTA strategy that prioritizes low computational complexity and rapid task assignment by selecting the nearest eligible drone after filtering based on availability and payload status. Unlike auction-based or consensus-based MRTA methods, which aim to optimize global task allocation, the proposed approach emphasizes low assignment latency for time-critical rescue operations. Future work will include a detailed comparative evaluation against classical MRTA algorithms.

Direct quantitative comparison across these frameworks is inherently limited, as prior works differ in disaster type, localization modality, validation methodology (simulation vs. physical deployment), and reported metrics. Table 14 positions SwarmSAR-FW against representative single-UAV and swarm-based SAR systems. Most swarm frameworks to date have been validated only in simulation, while single-UAV systems report detection or localization accuracy without coordination-latency or prioritization metrics. This comparison contextualizes SwarmSAR-FW's contributions rather than claiming strict superiority on every axis, given the differing experimental scopes across studies.

A limitation of the proposed framework is the assumption that wearable sensing devices provide victim physiological and positional information. In practical deployments, such devices may be distributed in advance to first responders, workers in high-risk environments, or residents of disaster-prone regions. Furthermore, many modern smartwatches already incorporate GPS, inertial measurement units, heart-rate sensors, wireless communication interfaces, and, in some cases, barometric sensors [47]. Consequently, the proposed wearable architecture can be integrated with commercially available wearable platforms through appropriate communication and data-sharing protocols. Importantly, the framework does not rely exclusively on wearable devices for victim detection, as onboard drone vision remains capable of detecting human presence independently. The wearable component primarily enhances victim prioritization, localization accuracy, and health-status assessment when such information is available. This work did not perform

TABLE 14. Comparison of SwarmSAR-FW with representative single-UAV and swarm-based SAR frameworks.

Framework	Methodology	Single/Swarm	Key Metrics	Limitations	Why SwarmSAR-FW Improves
SARDO [48] (2021)	Pseudo-trilateration and machine-learning-based localization of mobile phones using UAV-enabled cellular sensing	Single UAV	Localization accuracy of a few tens of meters; ∼3 min per user equipment; ∼5% battery consumption	Relies on cellular signal detection, not visual confirmation; no payload delivery; coarse (tens of meters) accuracy; no victim prioritization	SwarmSAR-FW achieves meter-level (1.6–1.8 m) visual localization and includes autonomous payload deployment, not just position estimation
Bio-inspired Thermal Swarm UAV [49] (2025)	Bio-inspired multi-UAV exploration using thermal sensing and optimization-based coordination in a PX4–Gazebo simulation	Swarm (simulated)	PSO exploration score of 0.67; 80% area coverage within 60% of mission time; redundancy ratio ∼0.25	Focused on thermal-based search and cooperative exploration in simulation; does not address physical rescue-payload delivery or flood-specific victim dynamics	SwarmSAR-FW combines cooperative search with visual victim/fire localization, victim prioritization, flood-specific drift-aware assistance, and physical multi-UAV validation.
Multi-UAV DRL Post-Disaster SAR [50] (2025)	Deep reinforcement learning for multi-UAV collaborative post-disaster search, using a linear-attention mechanism to reduce computational overhead in high-dimensional coordination	Swarm (algorithmic / simulation-based)	Evaluates multi-UAV collaborative search and computational efficiency of the proposed DRL approach	Focuses on autonomous multi-UAV search and decision-making; does not integrate physical rescue-payload deployment, victim prioritization, or fire-and-flood mission modes	SwarmSAR-FW combines multi-UAV coordination with victim prioritization, rescue-payload deployment, flood-specific victim tracking, and physical UAV experiments.
Drone Swarm SAR [23] (2025)	Swarm-based SAR framework evaluated in terms of coverage efficiency, victim detection, communication resilience, energy utilization, and adaptability	Swarm	42% faster surveying than single-UAV deployment; victim-detection accuracy approaching 90%; communication resilience above 85%; performance loss below 4% under adverse conditions	Demonstrates the broader benefits of swarm-based SAR but does not report integrated physical rescue-payload deployment, victim prioritization, or a unified fire-and-flood SAR framework	SwarmSAR-FW integrates victim detection with autonomous rescue-payload deployment, priority-based assistance, and separate fire- and flood-response mechanisms, with physical UAV validation.

formal stability analysis of the swarm under sustained communication delays. Nevertheless, the centralized Firebase coordination framework preserves task states during temporary communication interruptions, allowing drones to continue executing their most recently assigned tasks until synchronization is restored. While transient delays primarily increase coordination latency, prolonged communication disruptions may reduce coordination efficiency. Future work will include a rigorous stability analysis under varying communication delays.

Although the proposed framework demonstrated reliable end-to-end operation under controlled flight experiments, the reliance on injected visual scenes remains the primary limitation of this study. While the perception algorithms were evaluated using real image datasets and the autonomous flight, communication, swarm coordination, and rescue logic were validated using physical UAVs, the complete system has not yet been evaluated during genuine water-rescue or live-fire incidents. Future work will focus on collaborations with emergency response organizations to conduct controlled field evaluations using trained personnel, safety supervision, and authorized emergency training environments to further assess system robustness under realistic operational conditions.

The current experiments were conducted under wind speeds of 0–2 m/s with no precipitation. Heavy rain introduces additional operational challenges that may affect overall system performance and mission efficiency. Evaluation under precipitation conditions is identified as a necessary future step toward operational deployment. Nevertheless, the proposed framework is designed to maintain functionality under a range of environmental conditions through adaptive survey and coordination strategies. The current framework was not evaluated under deliberate adversarial sensing failures. However, the swarm coordination architecture provides partial resilience by allowing unaffected drones to continue operating independently, while the centralized victim reservation mechanism supports the reassignment of unresolved tasks. Comprehensive evaluation under adversarial sensing conditions remains an important direction for future research. Under partial swarm failure, if a drone becomes unavailable during mission execution, the remaining drones continue operating within their assigned regions. Tasks associated with the unavailable drone can be released from the centralized reservation mechanism and reassigned to other eligible drones, allowing mission execution to continue. Although partial swarm failures may increase overall mission completion time, they do not prevent continued rescue operations. Quantitative evaluation under multiple simultaneous drone failures is identified as future work.

Under dense smoke conditions, frame-level fire detection performance decreases due to partial occlusion of visible flame regions. Nevertheless, the proposed FireDensity

temporal accumulation integrates evidence across multiple survey passes, preserving region-level fire localization despite intermittent visibility. Extremely dense smoke remains a challenging operating condition for RGB-based sensing and represents an important direction for future investigation.

Additionally, the framework assumes reliable communication and full sensor availability across all swarm members. Formal analysis of swarm stability under sustained communication delays was not performed; however, the centralized Firebase coordination model inherently tolerates transient delays, as task states are preserved in the database and re-queried upon reconnection. Under partial swarm failure, if one drone becomes unresponsive, the remaining drones continue independent operation within their assigned regions, and the victim reservation mechanism ensures no assigned task is permanently blocked. Convergence is guaranteed in finite time for a fixed set of detected victims at a given decision epoch, as each successful assignment reduces the number of unassigned victims by one. Newly detected victims are incorporated into subsequent assignment cycles, enabling continuous adaptation to evolving disaster conditions. The proposed method can be viewed as a simplified MRTA strategy. Auction-based approaches are promising alternatives but were not evaluated in this work and are left for future study.

VI. CONCLUSION

This work presented SwarmSAR-FW, a drone-based swarm search-and-rescue framework designed for both fire and water-related disaster scenarios. By integrating adaptive flight behaviors, multi-modal perception, and cooperative swarm coordination, the system enables rapid victim detection, prioritization, and assistance under highly dynamic and hazardous conditions. The proposed architecture supports dual operational modes saving and analyzing allowing drones to flexibly alternate between direct victim assistance and large-area situational assessment in response to real-time mission demands.

Experimental evaluations demonstrate the effectiveness of the proposed framework across both disaster scenarios. In water-rescue experiments, the onboard detection pipeline identified victims within an average of 2.1 s, while parachute-assisted life-jacket deployment achieved a mean drop offset of 1.8 m under nominal conditions. The physiological-priority assessment mechanism achieved 98.7% agreement with ground-truth urgency rankings, confirming reliable victim prioritization. When swarm coordination was required, the system reassigned tasks within an average of 4.1 s, with assisting drones reaching victims in approximately 25.4 s. In fire-disaster experiments, the analyzing-mode localization algorithm achieved an average fire-region localization error of 1.6 m and successfully identified the most hazardous region in 96.1% of trials, while the entry/exit detection module achieved an overall accuracy of 91.0%.

A key strength of SwarmSAR-FW lies in its resilience to ambiguous and near-threshold conditions. Extensive stress-testing demonstrated stable system behavior even when urgency scores, fire intensities, or spatial cues were closely matched, ensuring that no critical victims or high-risk regions were deprioritized during mission execution. Future work will focus on enhancing deployment reliability under extreme environmental conditions. In particular, improvements to the parachute deployment and stabilization mechanism will further reduce payload drift during water-based rescues under stronger winds and surface currents.

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AI STATEMENT

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